

Credit, housing and family size

FOUR SAVED HOUSEHOLD CASES

Financing changes at held prices

The main comparison changes the financed share of a home purchase from 80% to 100% in the current working chain-13 population. A separate extension repeats the comparison when every household remains at the lowest productivity state.

80% vs 100% financed Two populations 17 standard views per case

October 3, 2026 | Prepared from saved native solution arrays and plotted diagnostics

Interpretation boundary: these are fixed-price household solutions. A 100% financed share removes the modeled purchase down payment; it does not make all borrowing unrestricted. The report does not claim market clearing, a recalibration, or a policy general equilibrium.

CONTENTS

Scope and definitions / Current productivity distributions / Permanent lowest-productivity extension / Numeric comparison tables / Sources and limitations / Seventeen four-case diagnostic views

How to read this report

Experiment matrix

Current productivity process	80%	100%	Main comparison	
Permanent lowest productivity	80%	100%	Separate extension	

FINANCED SHARE

The fraction of a home purchase financed by mortgage debt. The down payment uses 1 minus this share.

CHILDREN EVER BORN

A stock of births up to the model age. It differs from the number of children currently at home.

OWNERSHIP SHARE

The population mass choosing owner tenure divided by total mass at the age or group shown.

AGE

The start of a four-year model cell for the reproductive household member, not a single-year survey interview age.

FIXED PRICE

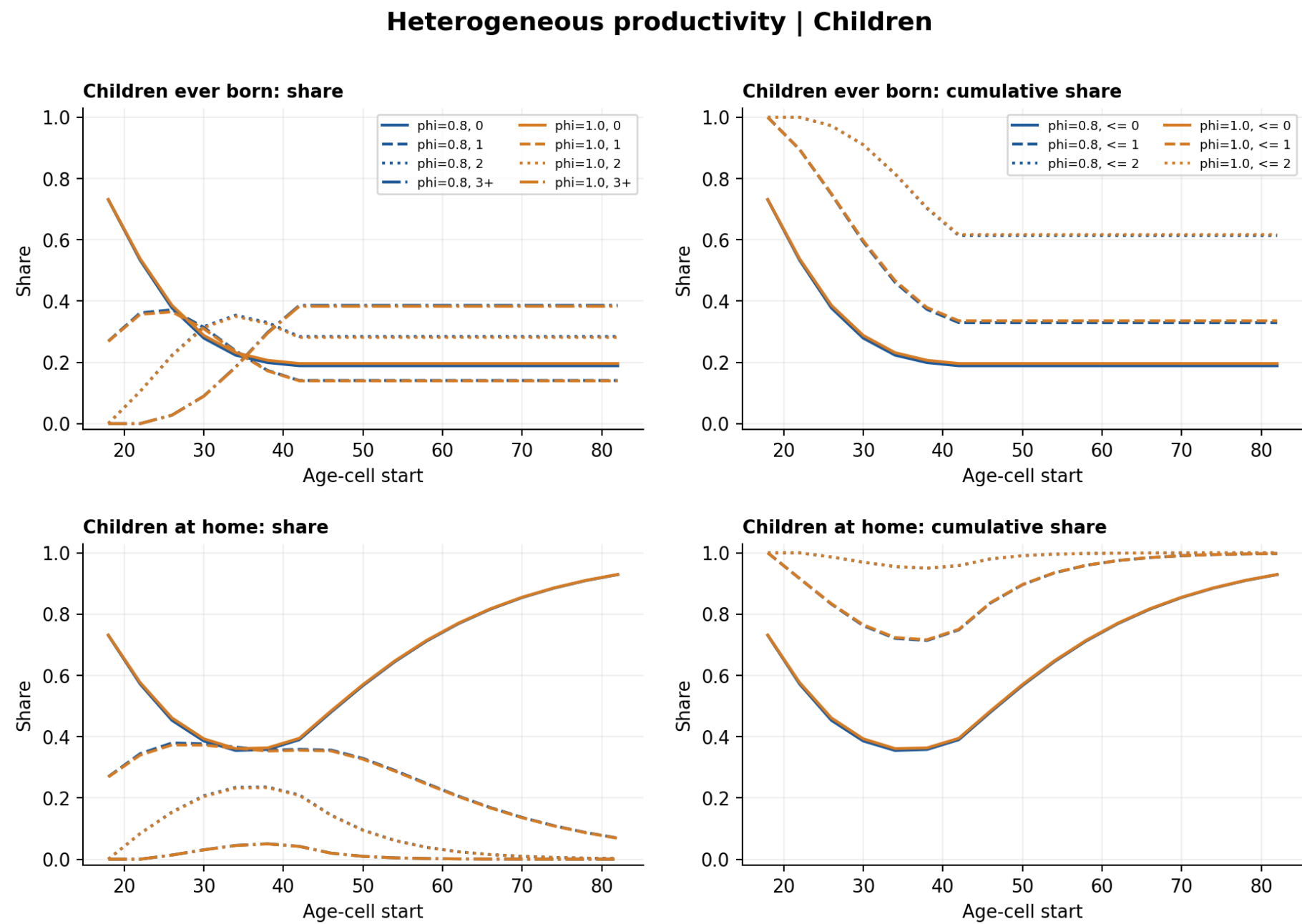
The saved house price is held while household choices and distributions change; housing markets need not clear.

The appendix preserves all 17 standard diagnostic plot types for each of the four saved cases. Raw policy lines can include unoccupied numerical income and wealth states. Unvalidated 100% financing consumption and next-asset aggregates are omitted from the main comparison.

Executed inputs: house price 0.776057 in all four cases. The lower-productivity extension sets permanent productivity to 0.103468, retains the entrant financial-wealth marginal, keeps the payroll-tax rate at 0.080281 and rebalances the four-year pension to 0.094961.

Current productivity process

Children ever born and currently at home

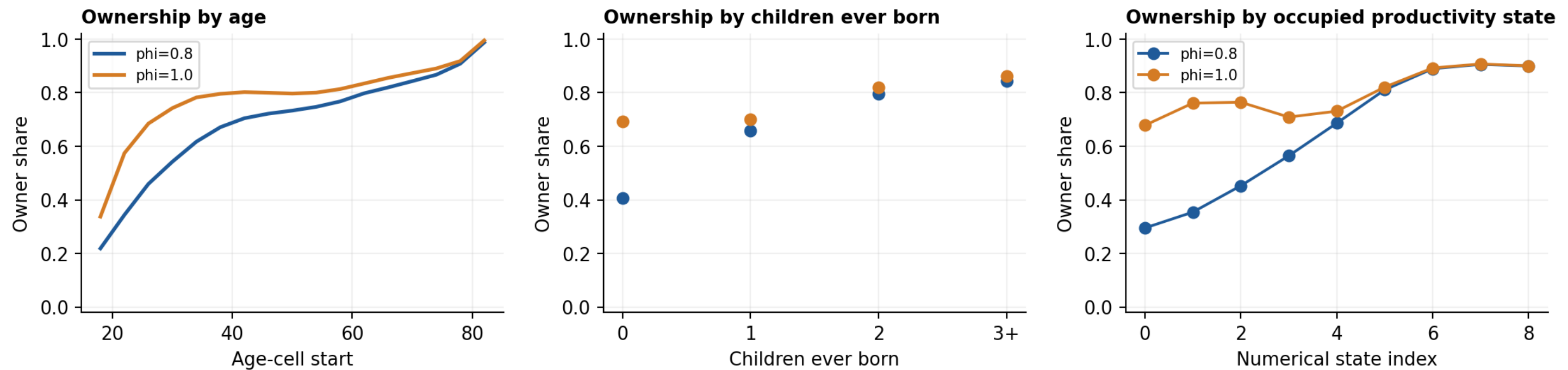


Children ever born and currently at home by four-year age cell. Each line divides by that case's own age-cell mass; 3+ is the top ever-born state.

Current productivity process

Ownership across age and family size

Heterogeneous productivity | Ownership

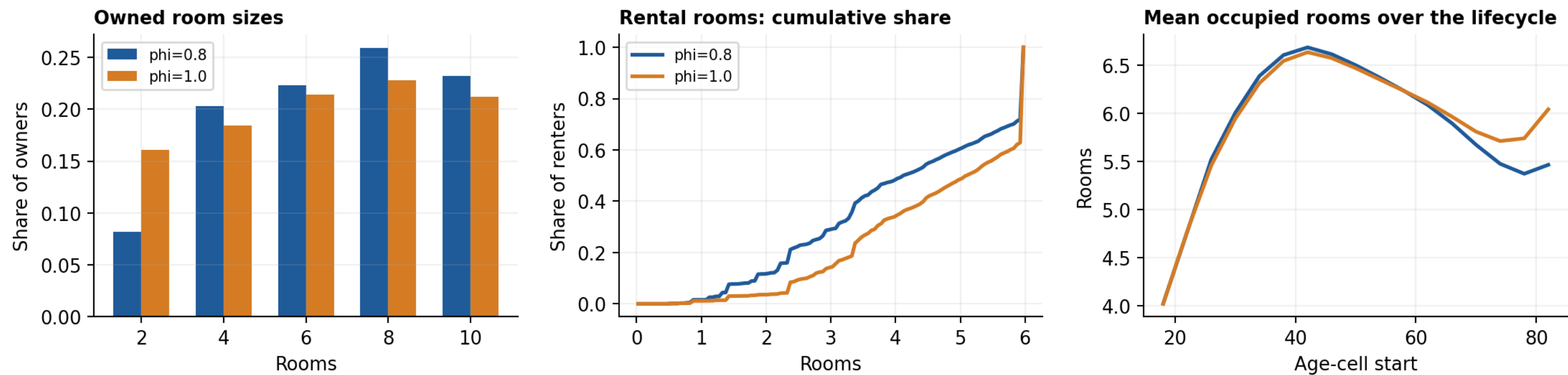


Ownership among all households by age, within child-count groups, and within occupied productivity states. Empty groups have no plotted rate.

Current productivity process

Housing services and room sizes

Heterogeneous productivity | Housing

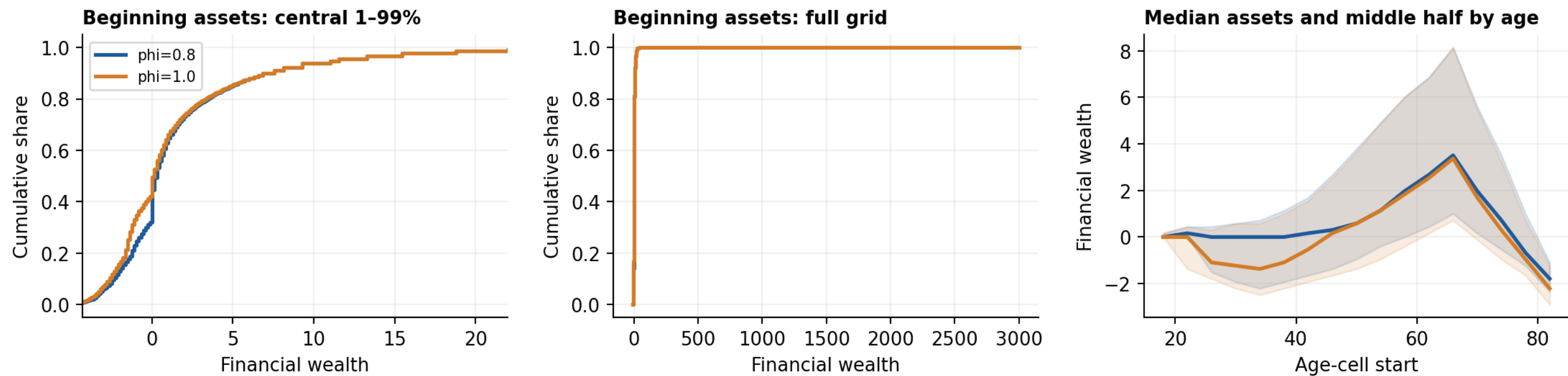


Owned rooms use realized owner tenure. Rental rooms use the saved conditional renter housing policy only on occupied renter mass after a finite and physical-range check.

Current productivity process

Beginning financial wealth

Heterogeneous productivity | Beginning financial wealth

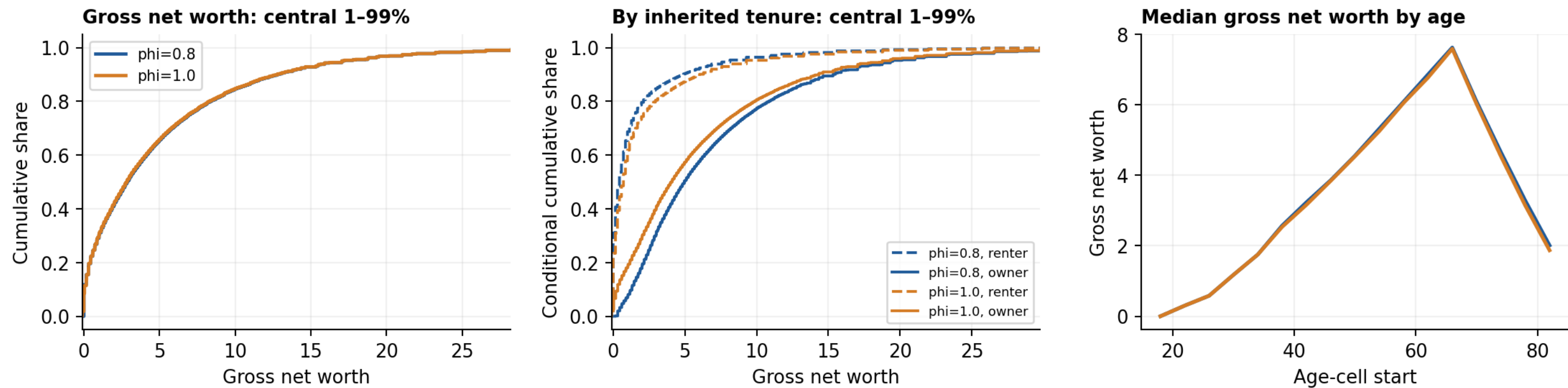


Financial wealth is beginning-of-period net financial assets using the post-fertility, pre-tenure distribution. Units retain the original annual gross-earnings normalization in both populations.

Current productivity process

Pre-transaction gross net worth

Heterogeneous productivity | Gross net worth

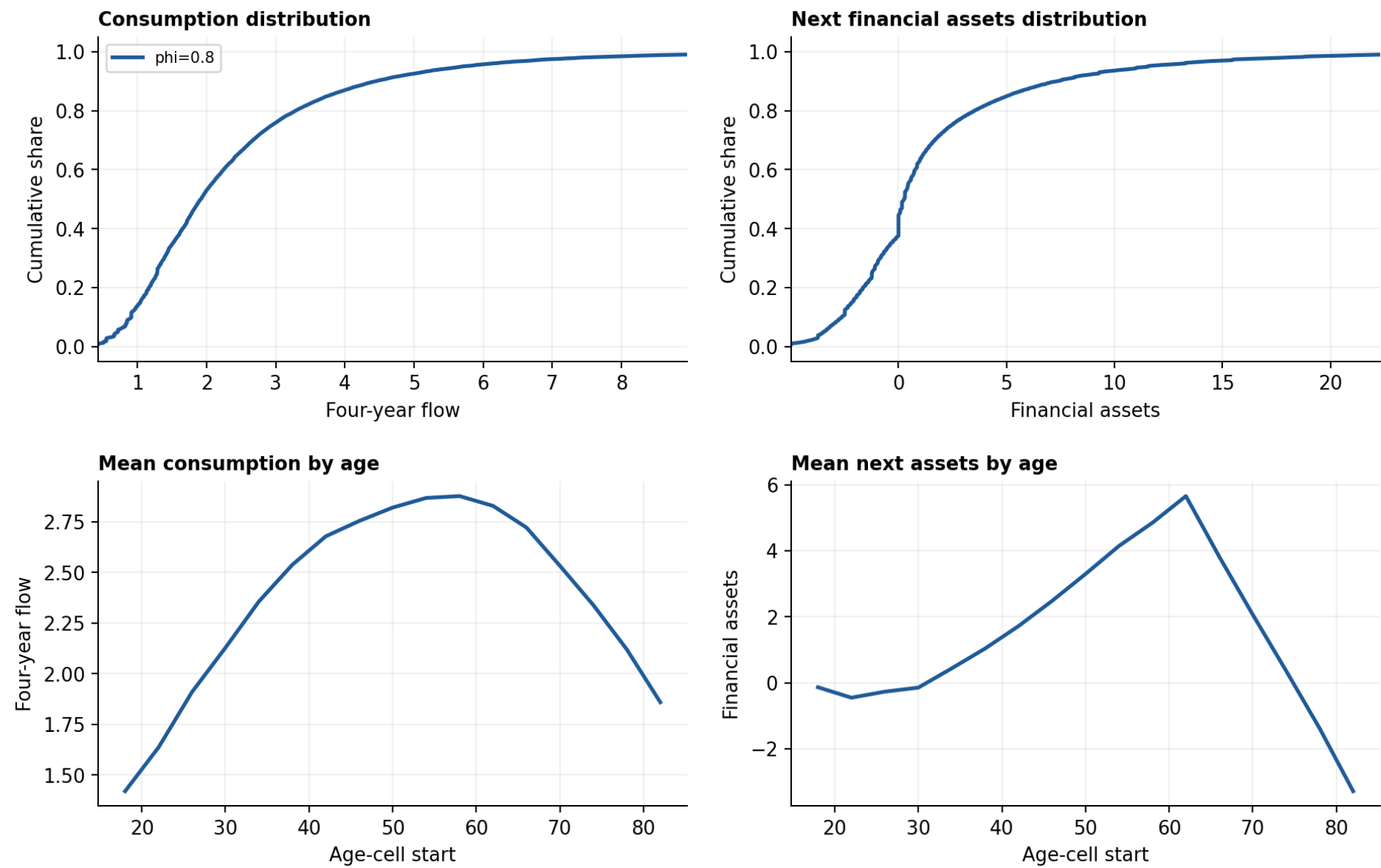


Pre-transaction gross net worth equals beginning net financial assets plus held house price times inherited owned rooms; renters have zero owned housing. CDF panels display each group's central 1–99% range without renormalizing probabilities; full tails are in the companion CSV. It excludes selling costs and is not an empirical wealth-target observer.

Current productivity process

Validated consumption and next-asset policies

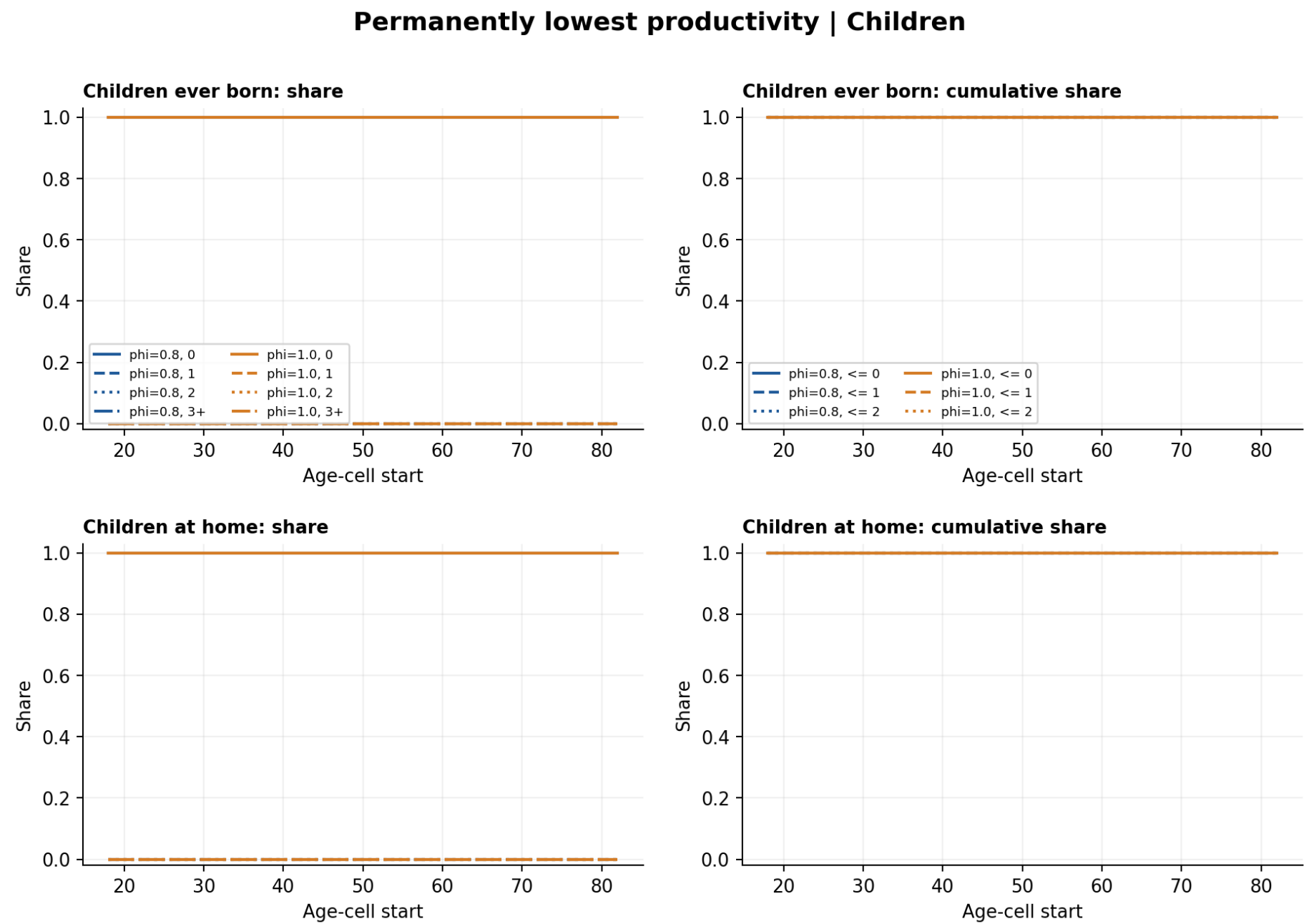
Heterogeneous productivity | Validated policies



Only the $\phi=0.8$ aggregate policy reporter passes its native value-function gate. $\phi=1$ is unavailable because positive buyer mass occupies initialized-infeasible value entries; no mass is filtered. Consumption is a four-year flow and next financial assets are a stock, not national-account saving.

Permanent lowest productivity

Children ever born and currently at home

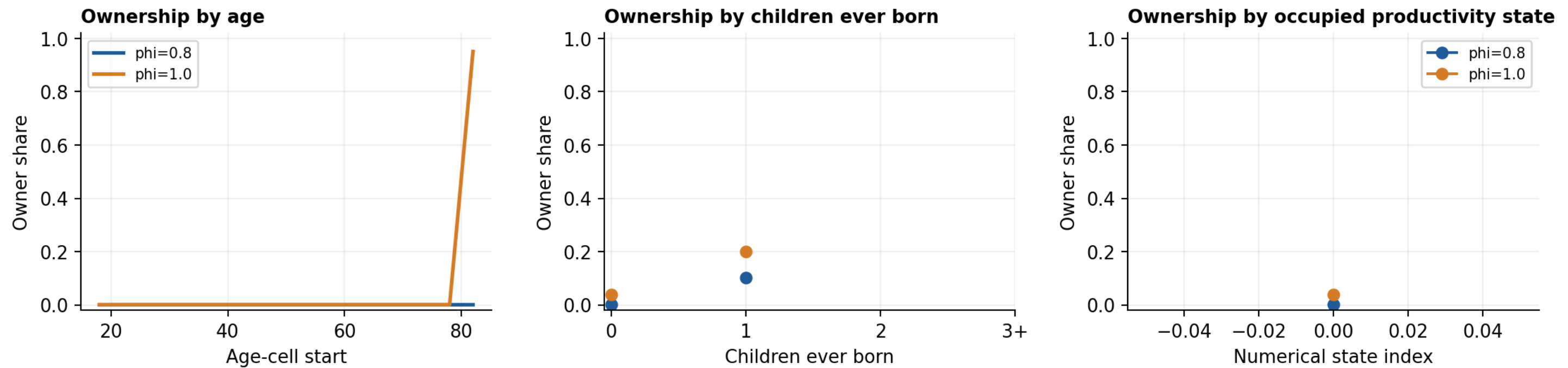


Children ever born and currently at home by four-year age cell. Each line divides by that case's own age-cell mass; 3+ is the top ever-born state.

Permanent lowest productivity

Ownership across age and family size

Permanently lowest productivity | Ownership

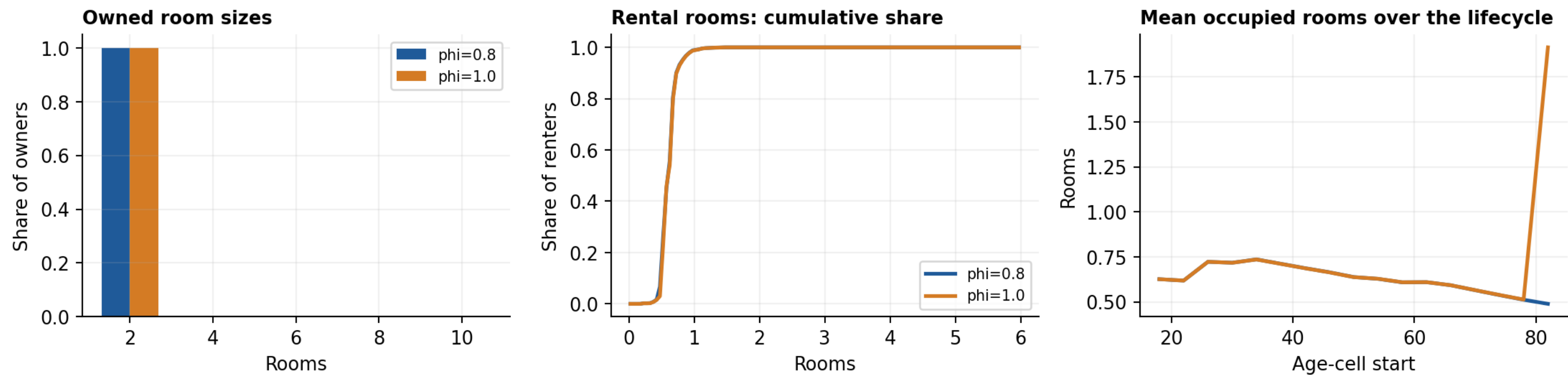


Ownership among all households by age, within child-count groups, and within occupied productivity states. Empty groups have no plotted rate.

Permanent lowest productivity

Housing services and room sizes

Permanently lowest productivity | Housing

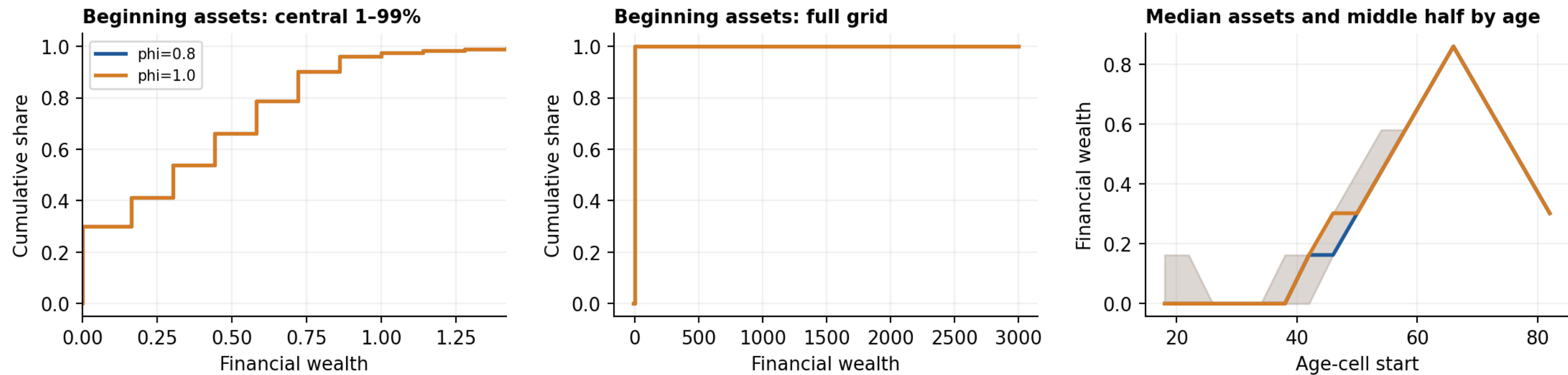


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Permanent lowest productivity

Beginning financial wealth

Permanently lowest productivity | Beginning financial wealth

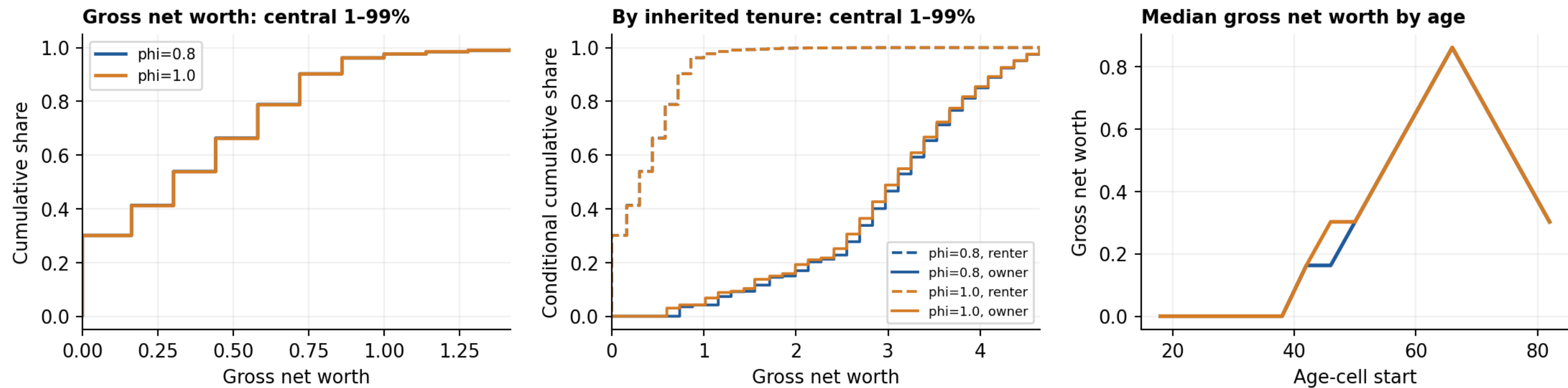


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Permanent lowest productivity

Pre-transaction gross net worth

Permanently lowest productivity | Gross net worth

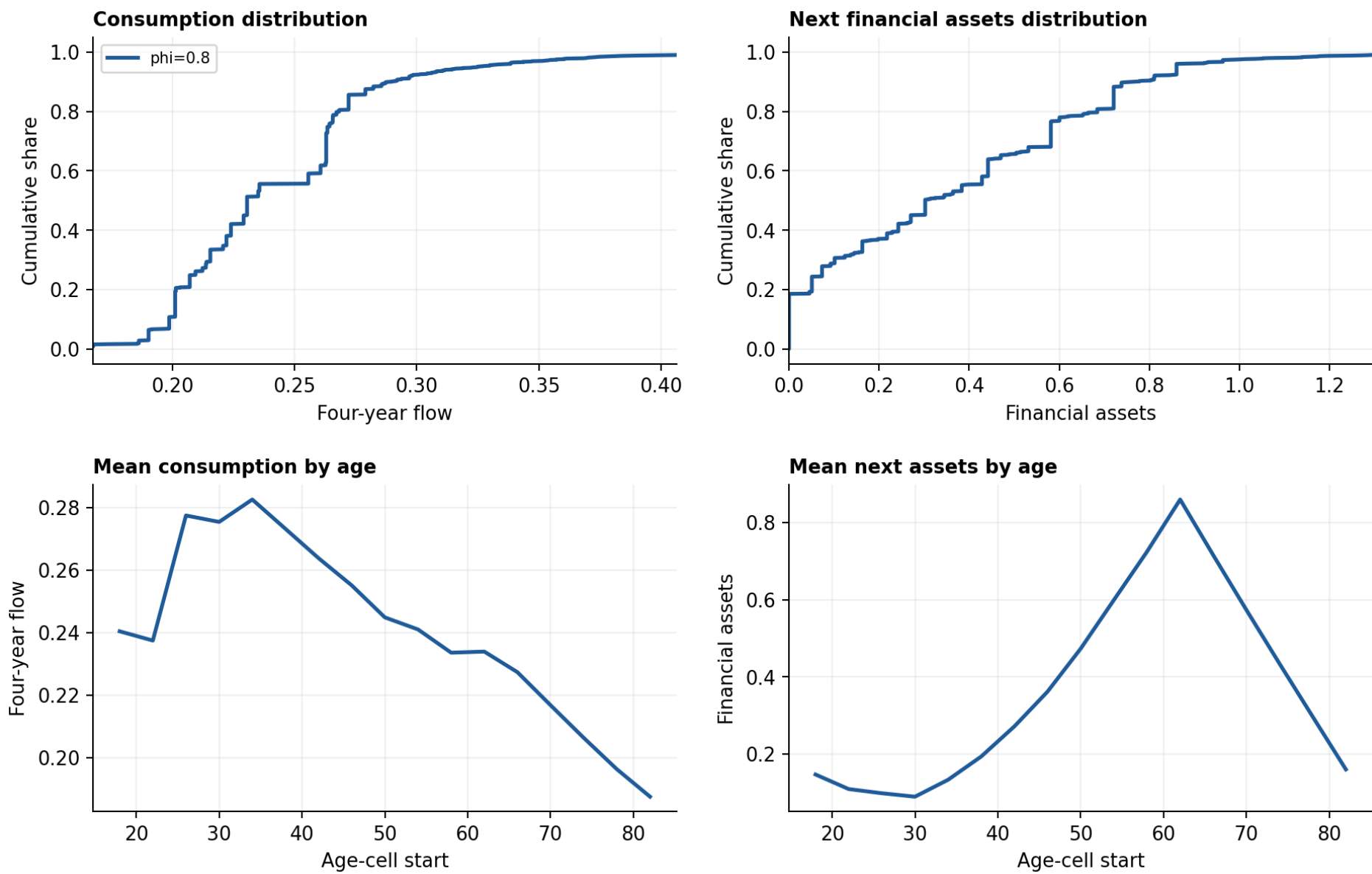


Pre-transaction gross net worth equals beginning net financial assets plus held house price times inherited owned rooms; renters have zero owned housing. CDF panels display each group's central 1–99% range without renormalizing probabilities; full tails are in the companion CSV. It excludes selling costs and is not an empirical wealth-target observer.

Permanent lowest productivity

Validated consumption and next-asset policies

Permanently lowest productivity | Validated policies



Only the $\phi=0.8$ aggregate policy reporter passes its native value-function gate. $\phi=1$ is unavailable because positive buyer mass occupies initialized-infeasible value entries; no mass is filtered. Consumption is a four-year flow and next financial assets are a stock, not national-account saving.

What the saved cases establish

CURRENT PRODUCTIVITY PROCESS

The all-age ownership share rises from 67.927% under 80% financing to 77.404% under 100% financing. At ages 42-45, the childless share is 18.888% and 19.611%, respectively. Credit relaxation changes ownership substantially while the children-ever-born comparison is smaller and has a different sign. These are saved household distributions at the same housing price.

PERMANENT LOWEST PRODUCTIVITY

The all-age ownership share rises from 0.077% to 3.792%, while age-22, age-30 and age-42 ownership remains near 0.08% in both financing arms. Virtually all households have no children ever born at every saved age; the largest share with any children is 1.73e-12. Productivity and the balanced pension also differ from the current population.

NUMERICAL QUALIFICATION

Both 100% financed native distributions pass their distribution checks, but the unchanged aggregate policy reporter stops at infeasible buyer-value cells. The flagged population masses are 5.26e-38 in the current-productivity case and 2.68e-15 in the lower-productivity case. Their consumption and next-asset aggregates are therefore unavailable. The raw distribution still supports the children, ownership, room and financial-wealth figures.

ECONOMIC SCOPE

The four cases retain the working post-interest chain-13 parameters, current birth menu, no Estate A and fixed house prices. There is no price adjustment, new calibration, general-equilibrium claim or population forecast.

Four-case outcome overview

Source: output/model/experiments/credit_distribution_report_v1/tables/comparison_overview.csv. More granular age, wealth and room tables are retained as companion CSVs in the same analysis folder.

case	population	phi	price held	ownership all ages	childless share age42	two plus share age42	owner mean rooms	renter mean rooms	policy reporter
reference_phi08	reference	phi=0.8	0.776057	0.679274	0.18888	0.670436	6.71436	4.13285	validated
reference_phi10	reference	phi=1.0	0.776057	0.774038	0.196107	0.664312	6.29286	4.67989	Unavailable
low_phi08	low	phi=0.8	0.776057	0.000770688	1	0	2	0.63217	validated
low_phi10	low	phi=1.0	0.776057	0.0379234	1	0	2	0.637258	Unavailable

Age-cell distribution summary: selected ages

Source: output/model/experiments/credit_distribution_report_v1/tables/comparison_selected_ages.csv. More granular age, wealth and room tables are retained as companion CSVs in the same analysis folder.

case	age cell	ownership	mean rooms	children 0	children 1	children 2	children 3plus	children at home mean
reference_phi08	18-21	0.218835	4.02177	0.730887	0.269113	0	0	0.269113
reference_phi08	30-33	0.542486	6.00917	0.279055	0.31409	0.316968	0.0898864	0.882424
reference_phi08	42-45	0.70458	6.68902	0.18888	0.140684	0.284477	0.385959	0.903354
reference_phi08	62-65	0.797863	6.08767	0.18888	0.140684	0.284477	0.385959	0.25712
reference_phi08	82-85	0.987391	5.46673	0.18888	0.140684	0.284477	0.385959	0.0731835
reference_phi10	18-21	0.337581	4.01982	0.731814	0.268186	0	0	0.268186
reference_phi10	30-33	0.742503	5.94888	0.287867	0.30954	0.313244	0.0893488	0.872261
reference_phi10	42-45	0.802045	6.6384	0.196107	0.139581	0.281335	0.382978	0.896626
reference_phi10	62-65	0.834354	6.11437	0.196107	0.139581	0.281335	0.382978	0.255205
reference_phi10	82-85	0.994982	6.04432	0.196107	0.139581	0.281335	0.382978	0.0726384
low_phi08	18-21	0.000643611	0.626133	1	0	0	0	0
low_phi08	30-33	0.00078335	0.717355	1	0	0	0	0
low_phi08	42-45	0.00078231	0.687319	1	1.11036e-12	0	0	1.11036e-12
low_phi08	62-65	0.000779786	0.609719	1	1.09676e-12	0	0	3.16039e-13
low_phi08	82-85	0.000776672	0.489223	1	1.07646e-12	0	0	8.99537e-14
low_phi10	18-21	0.000643113	0.62613	1	0	0	0	0
low_phi10	30-33	0.00078337	0.717357	1	0	0	0	0
low_phi10	42-45	0.000782338	0.686815	1	1.73303e-12	0	0	1.73303e-12
low_phi10	62-65	0.000781983	0.609711	1	1.72069e-12	0	0	4.93269e-13
low_phi10	82-85	0.950573	1.91335	1	1.7047e-12	0	0	1.40398e-13

Owned room-size shares

Source: output/model/experiments/credit_distribution_report_v1/tables/comparison_owner_rooms.csv. More granular age, wealth and room tables are retained as companion CSVs in the same analysis folder.

case	rooms	share among owners
reference_phi08	2	0.0819402
reference_phi08	4	0.203023
reference_phi08	6	0.223405
reference_phi08	8	0.259184
reference_phi08	10	0.232448
reference_phi10	2	0.160663
reference_phi10	4	0.184637
reference_phi10	6	0.214469
reference_phi10	8	0.228069
reference_phi10	10	0.212162
low_phi08	2	1
low_phi08	4	8.24031e-12
low_phi08	6	0
low_phi08	8	0
low_phi08	10	0
low_phi10	2	1
low_phi10	4	4.96049e-13
low_phi10	6	0
low_phi10	8	0
low_phi10	10	0

Financial-asset and validated policy quantiles

Source: output/model/experiments/credit_distribution_report_v1/tables/comparison_asset_quantiles.csv. More granular age, wealth and room tables are retained as companion CSVs in the same analysis folder.

case	beginning assets q10	beginning assets q50	beginning assets q90	gross net worth q50	consumption q50	next assets q50	policy reporter
reference_phi08	-2.34884	0.302326	7.57143	2.86564	1.91721	0.26051	validated
reference_phi10	-2.62791	0.162791	6.86047	2.84841	n/a	n/a	Unavailable
low_phi08	0	0.302326	0.72093	0.302326	0.230477	0.302326	validated
low_phi10	0	0.302326	0.72093	0.302326	n/a	n/a	Unavailable

Sources, reproduction and limits

SAVED CASE IDENTITIES

Main reference: output/model/fixed_reference_economics_20260928/soft_timing_calibration_20261002_v1/fable_analysis/credit_mechanism/credit_relaxation/. Lower productivity: output/model/experiments/low_productivity_credit_v1/. Each case retains its executed parameters, solution arrays and standard diagnostics.

NO NEW SOLVE

The accompanying output/model/experiments/credit_distribution_report_v1/manifest.json records each included figure and table. Granular CSVs remain beside the figures. Regenerate this PDF with build_pdf.py in the same folder, using the bundled Python environment. Figure and table preparation reads saved arrays only.

COMPARABILITY

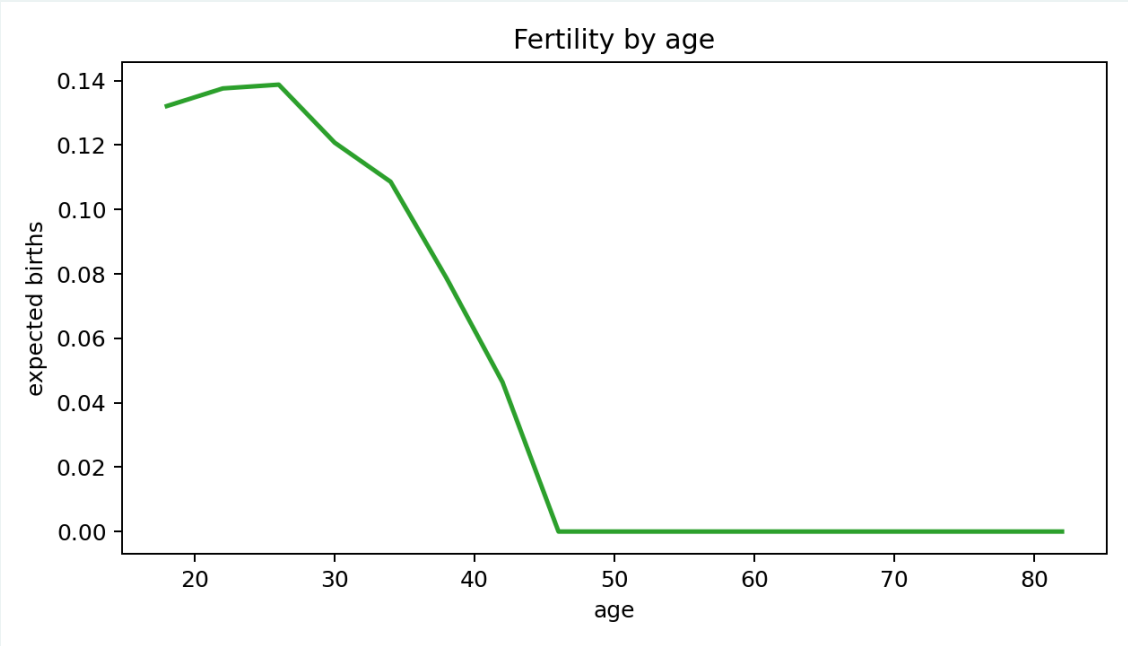
Within each population, the 80% and 100% financing arms hold the other executed household inputs and the house price fixed. Across populations, productivity, entrant wealth-income allocation and balanced pension differ. The latter comparison therefore cannot be attributed to financing alone.

RAW DIAGNOSTIC SCOPE

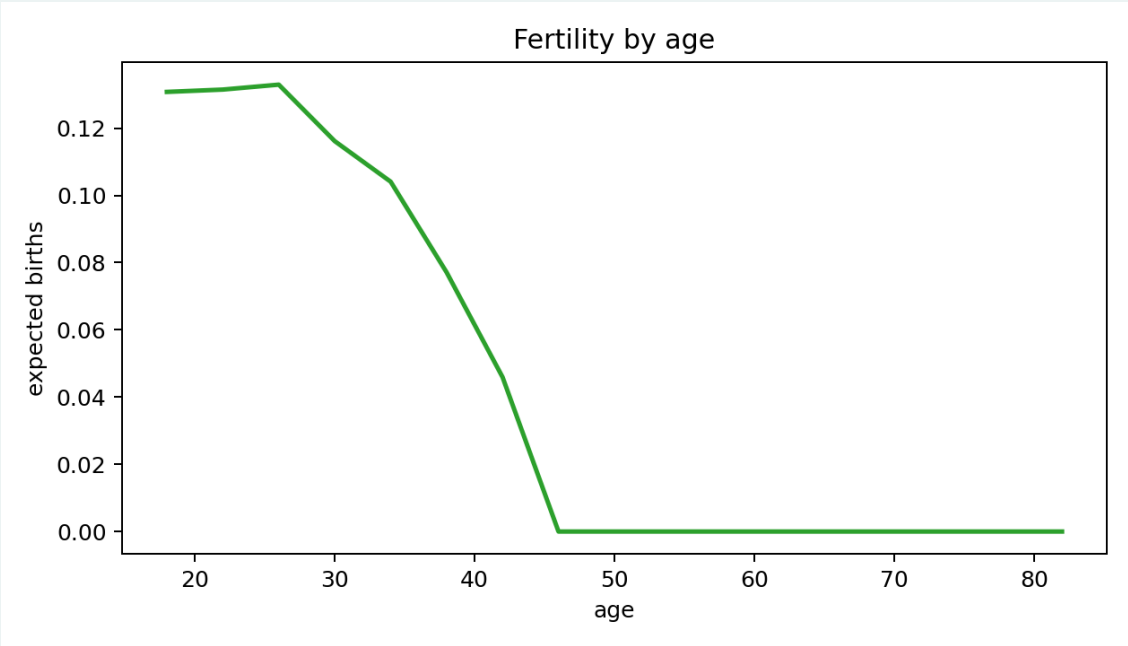
The appendix reproduces the standard diagnostic images for auditability. Several plot titles come from a general-equilibrium diagnostic routine, but their market residual panels here are evaluations at held prices. Income-state policy lines in the lower-productivity population include eight unoccupied numerical nodes.

Children and births by age

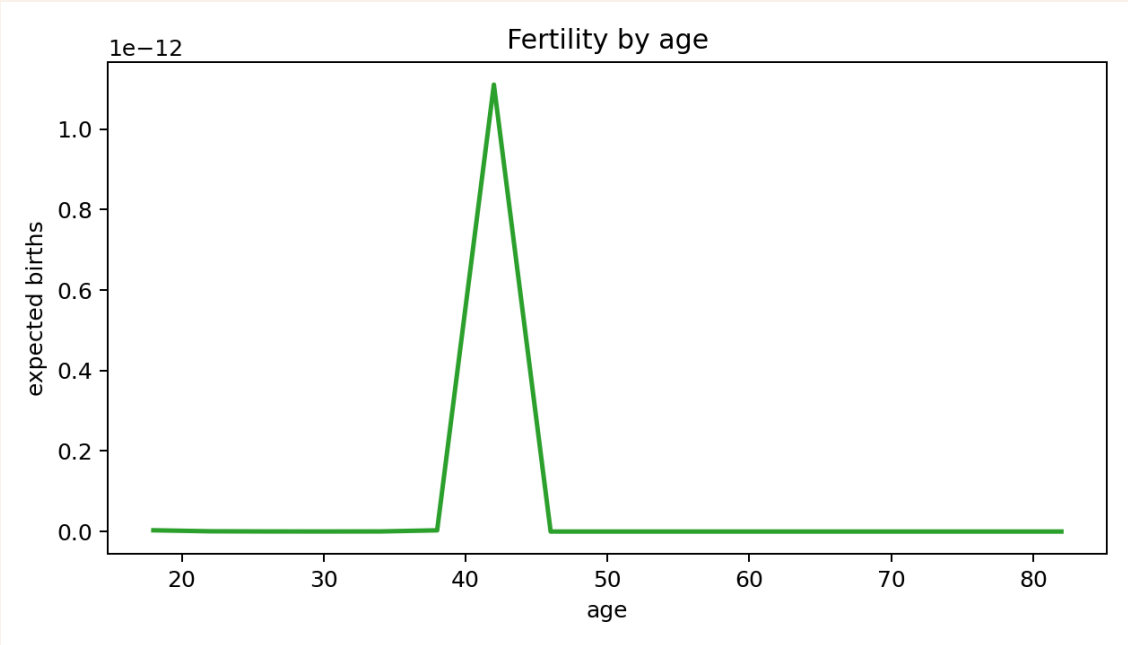
Current productivity / 80% financed



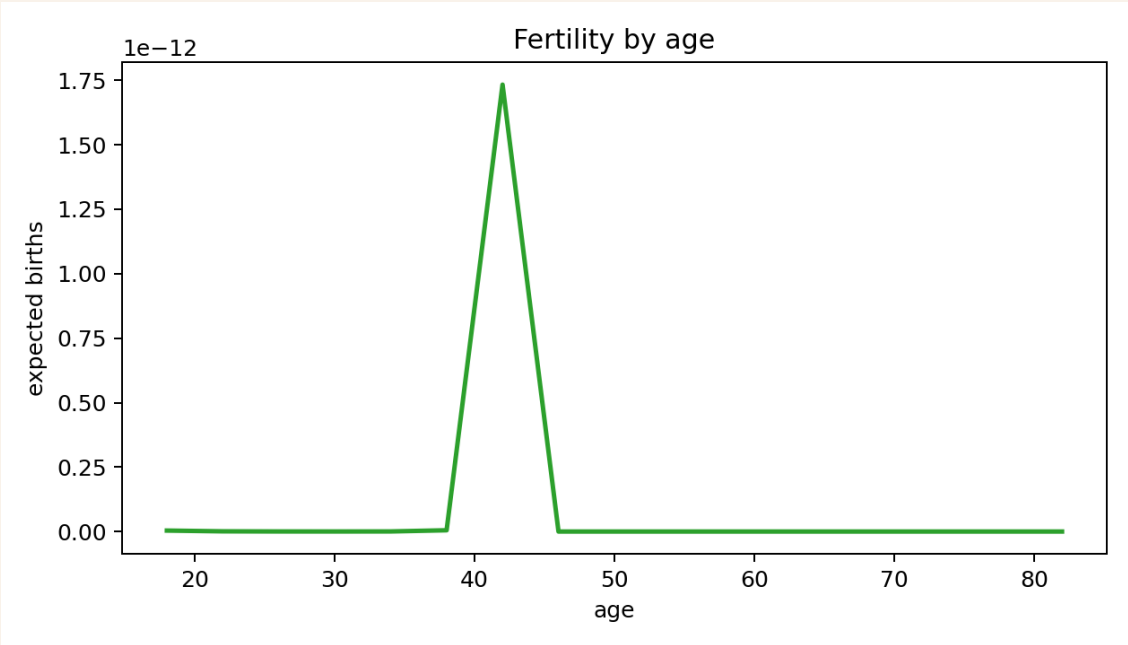
Current productivity / 100% financed



Lowest productivity / 80% financed



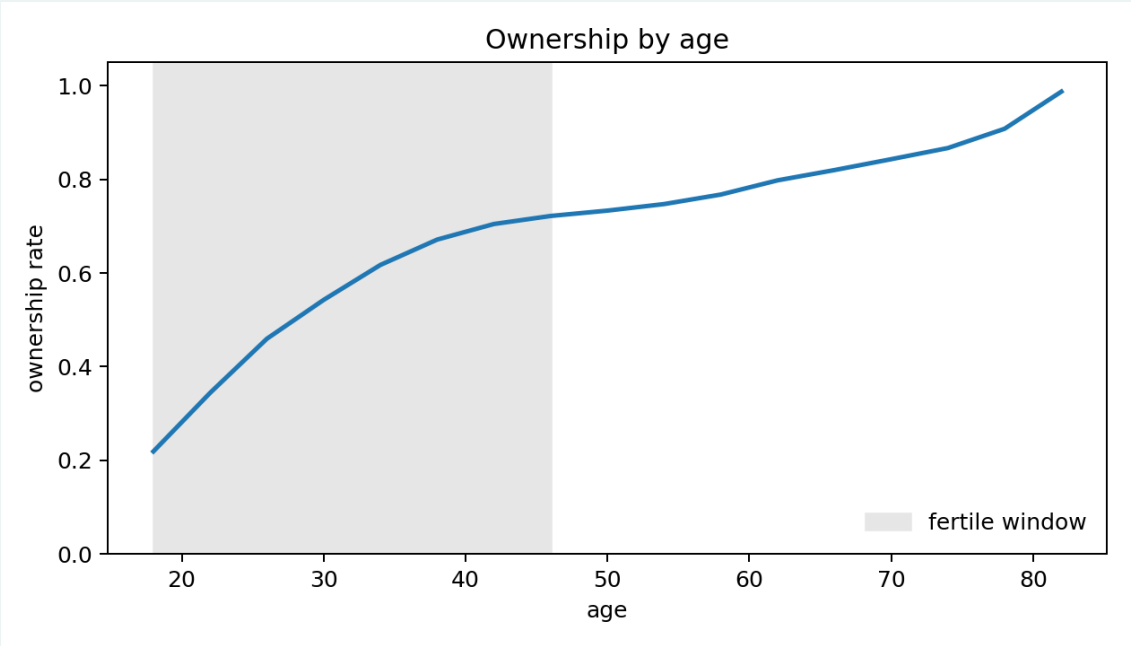
Lowest productivity / 100% financed



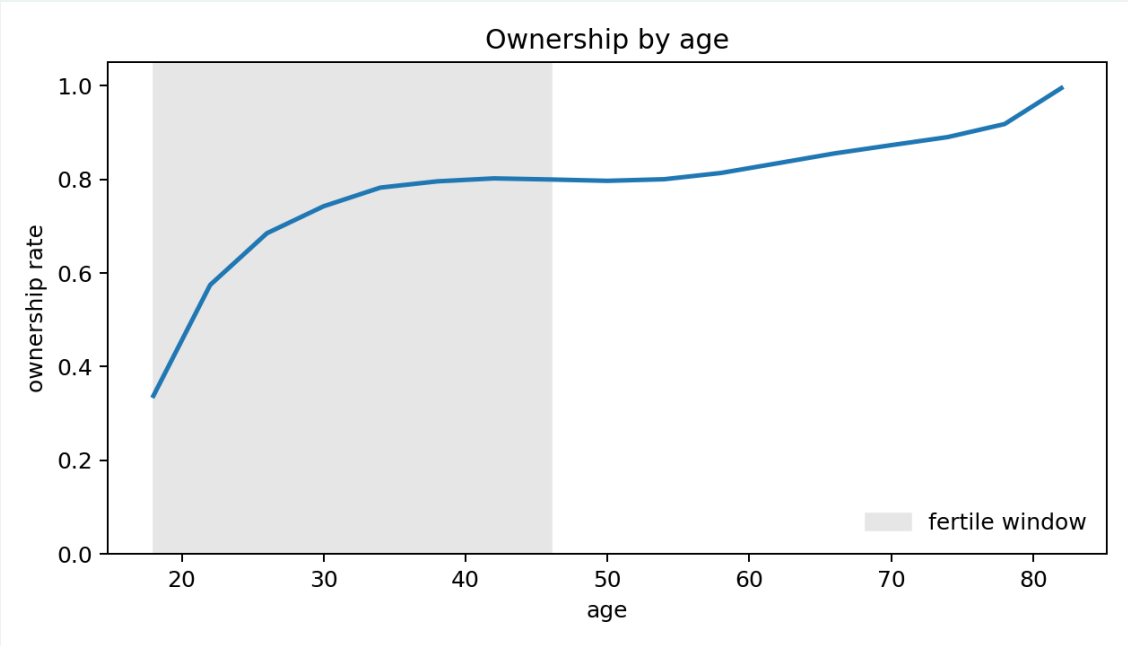
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Homeownership by age

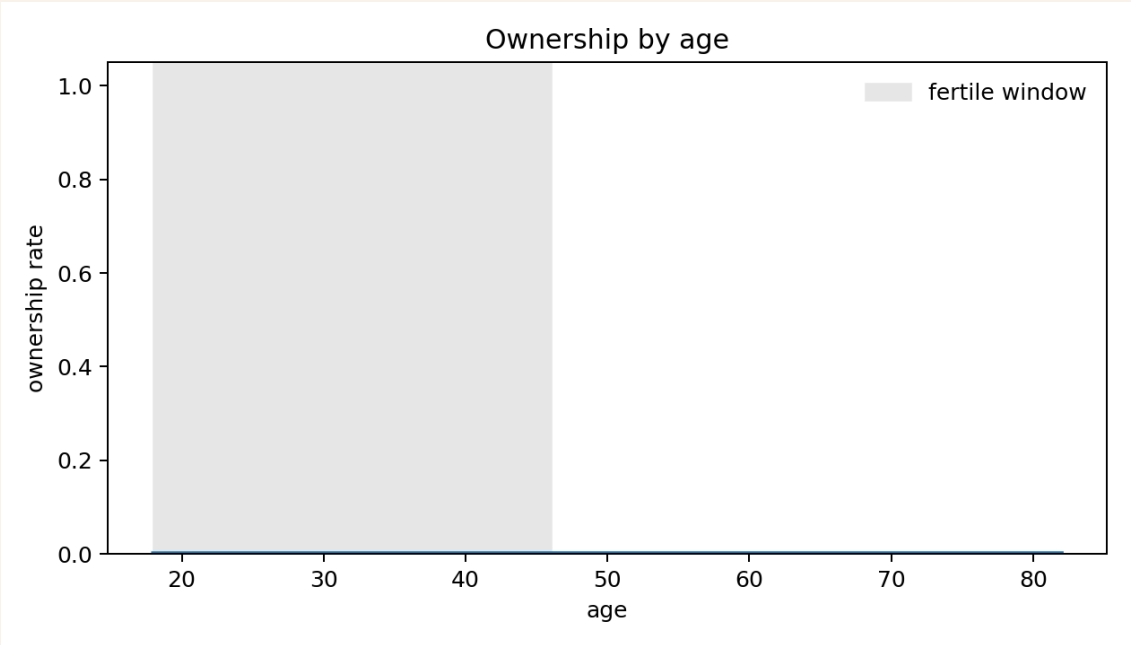
Current productivity / 80% financed



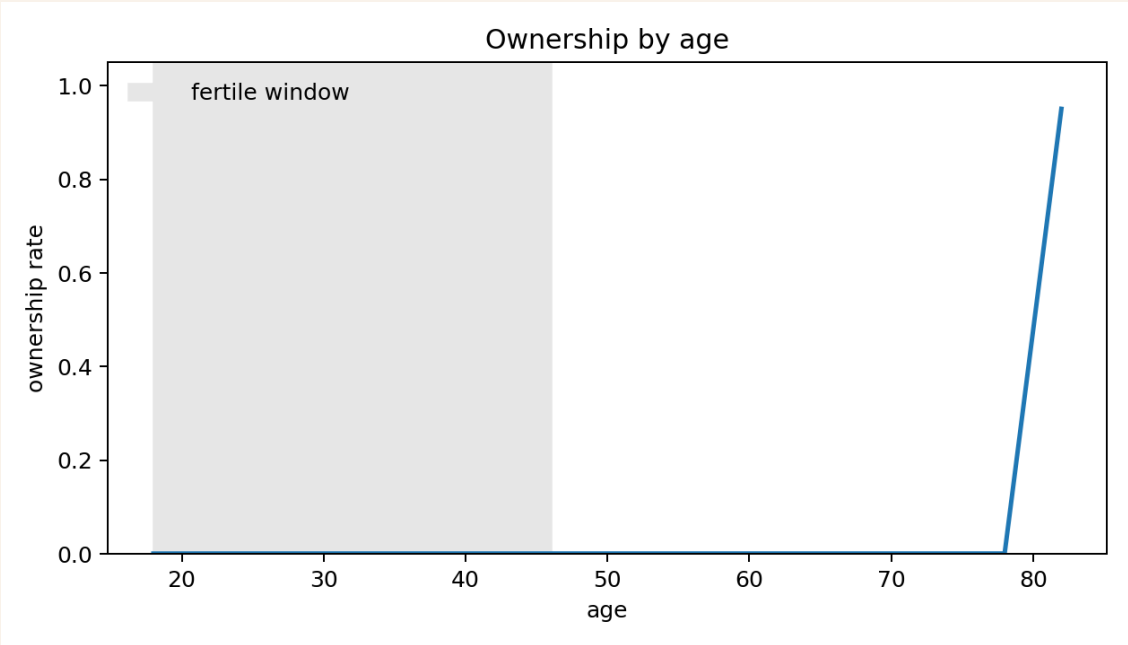
Current productivity / 100% financed



Lowest productivity / 80% financed



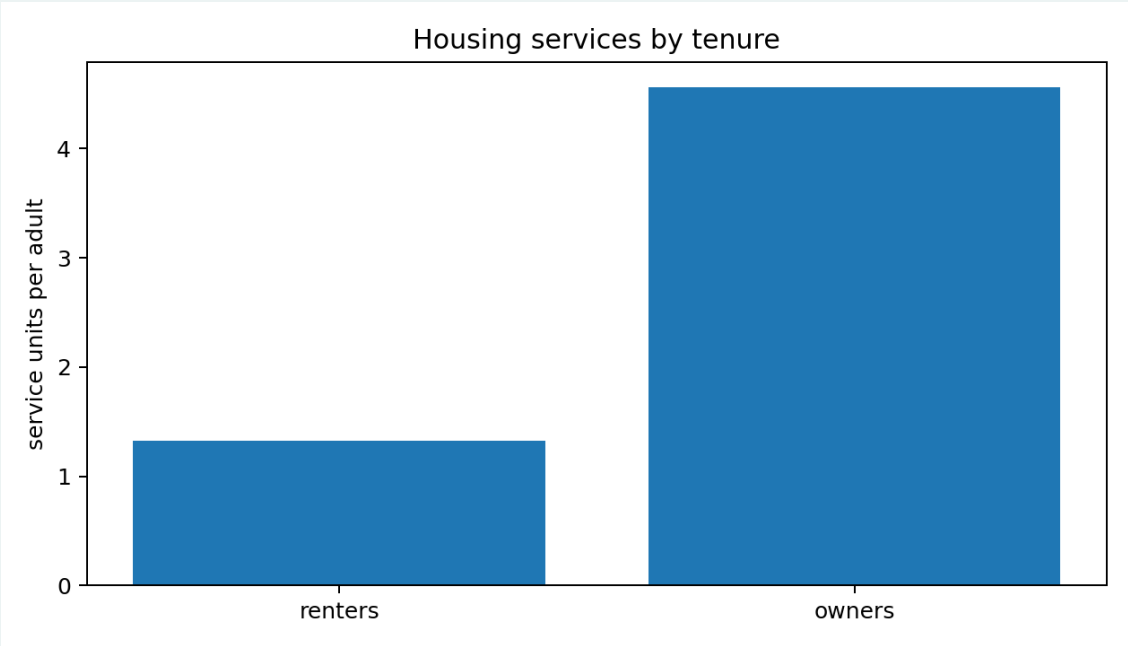
Lowest productivity / 100% financed



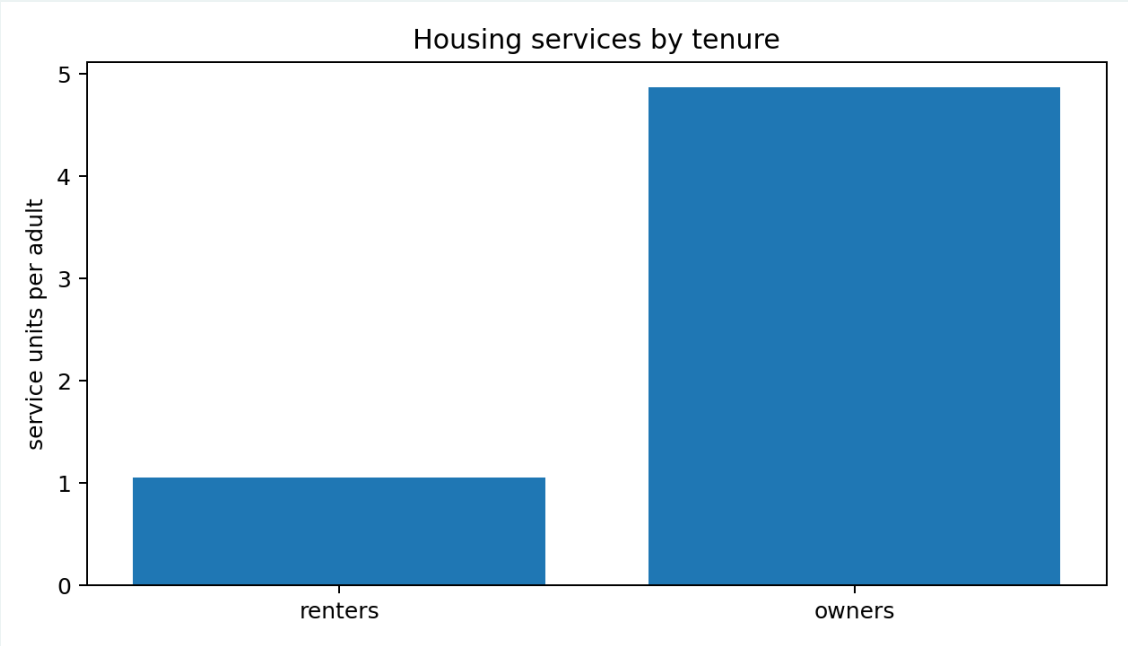
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Tenure and housing services

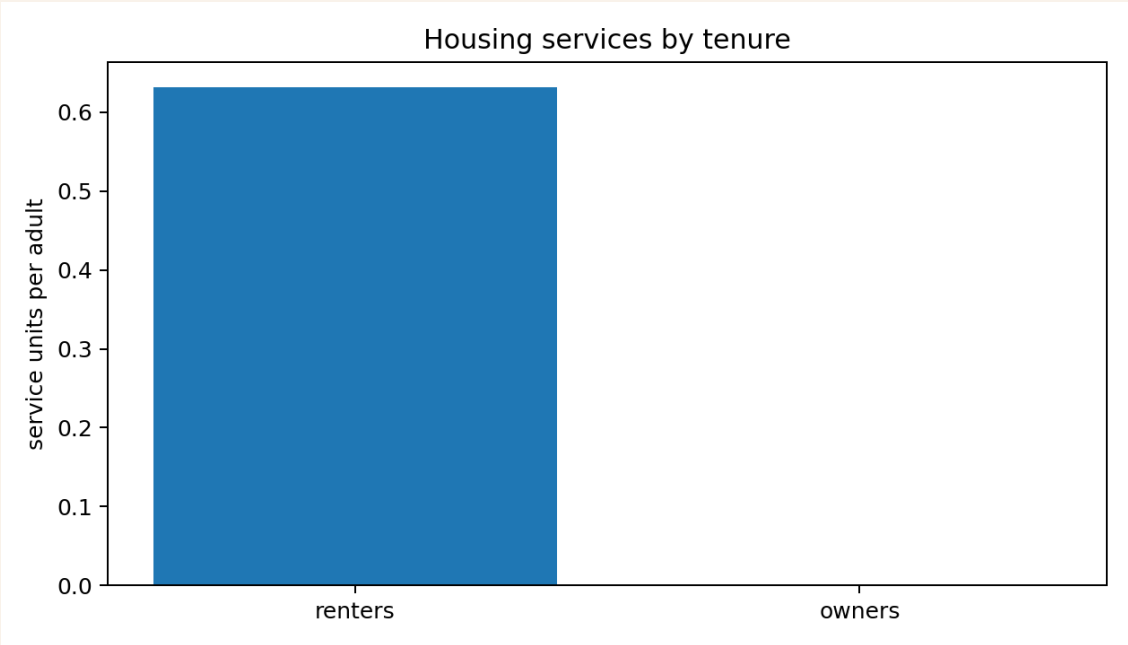
Current productivity / 80% financed



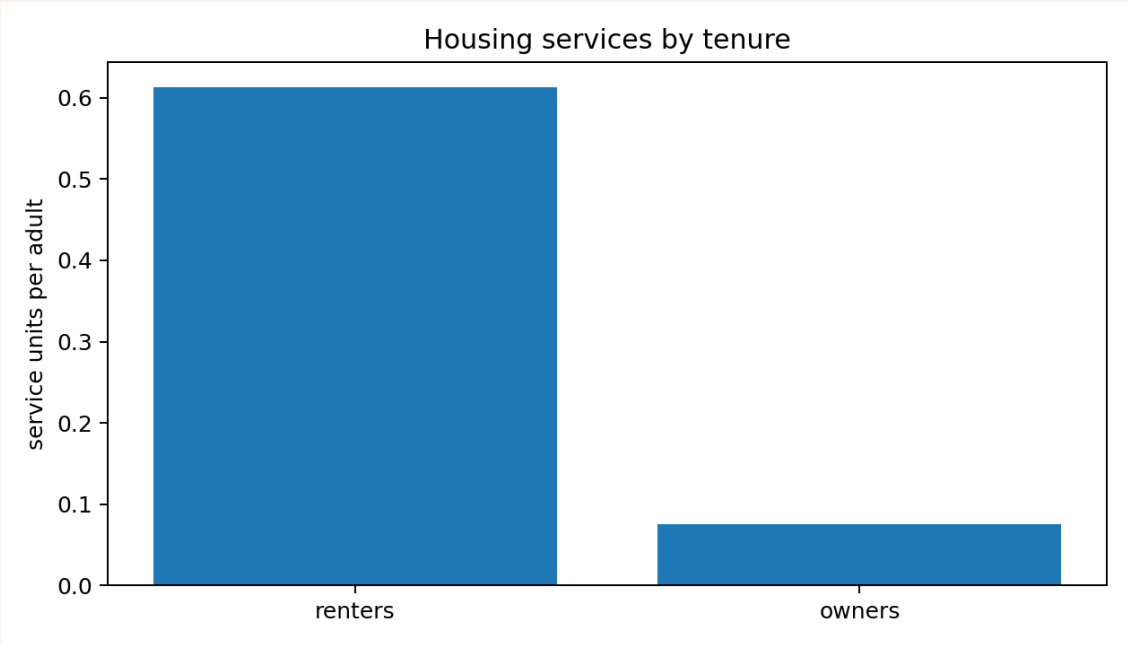
Current productivity / 100% financed



Lowest productivity / 80% financed



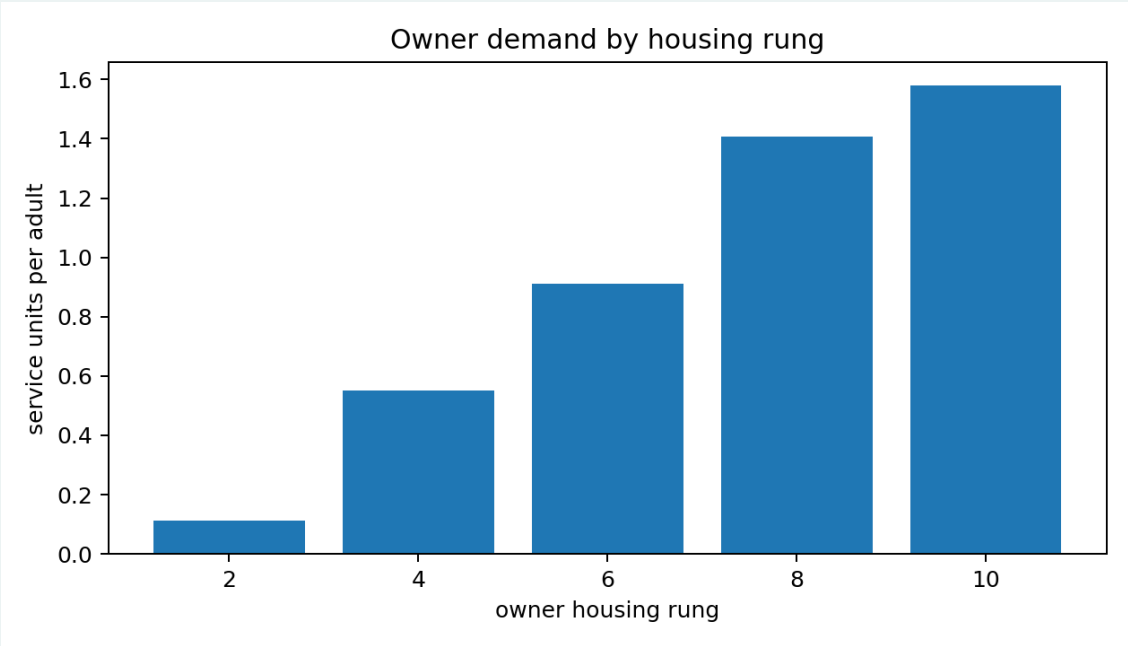
Lowest productivity / 100% financed



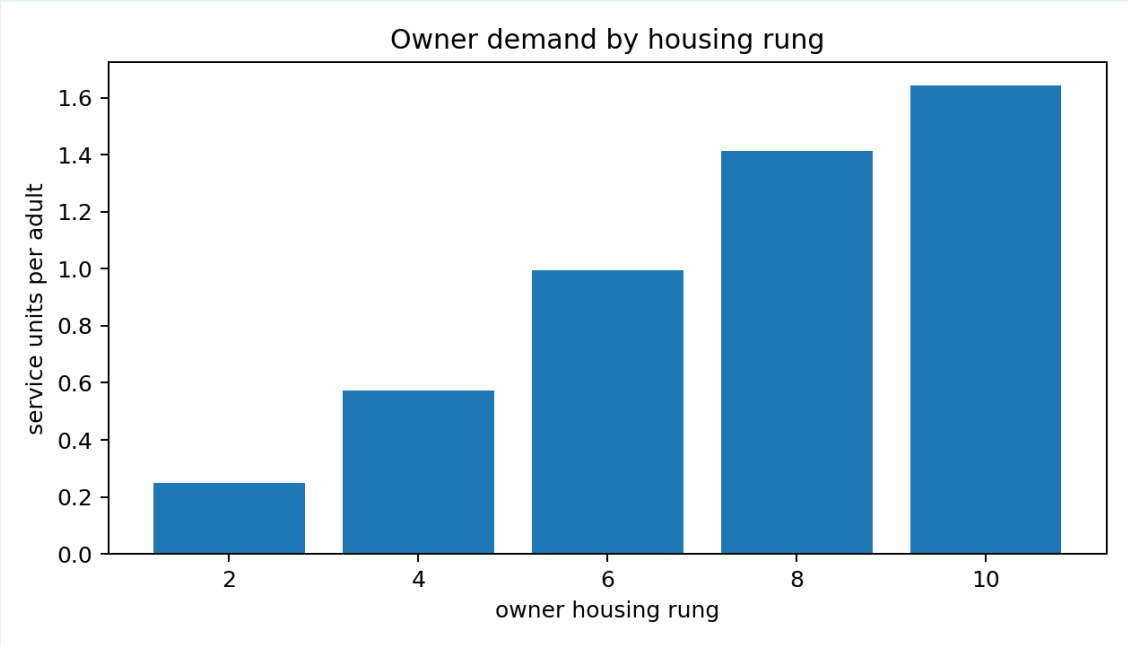
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Owner housing products

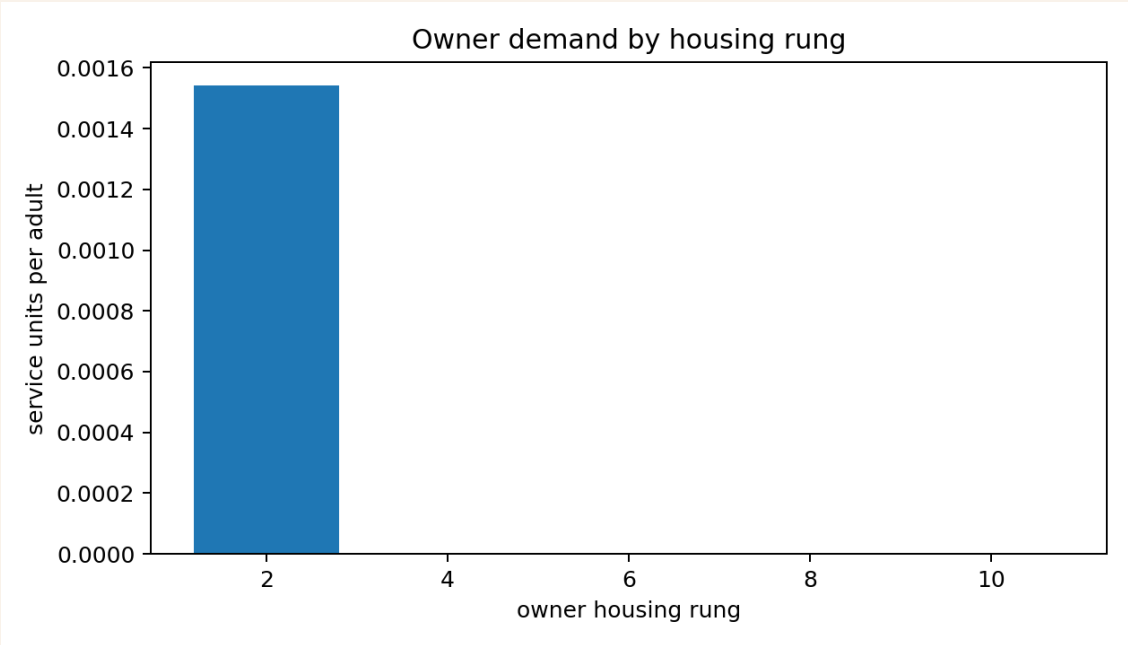
Current productivity / 80% financed



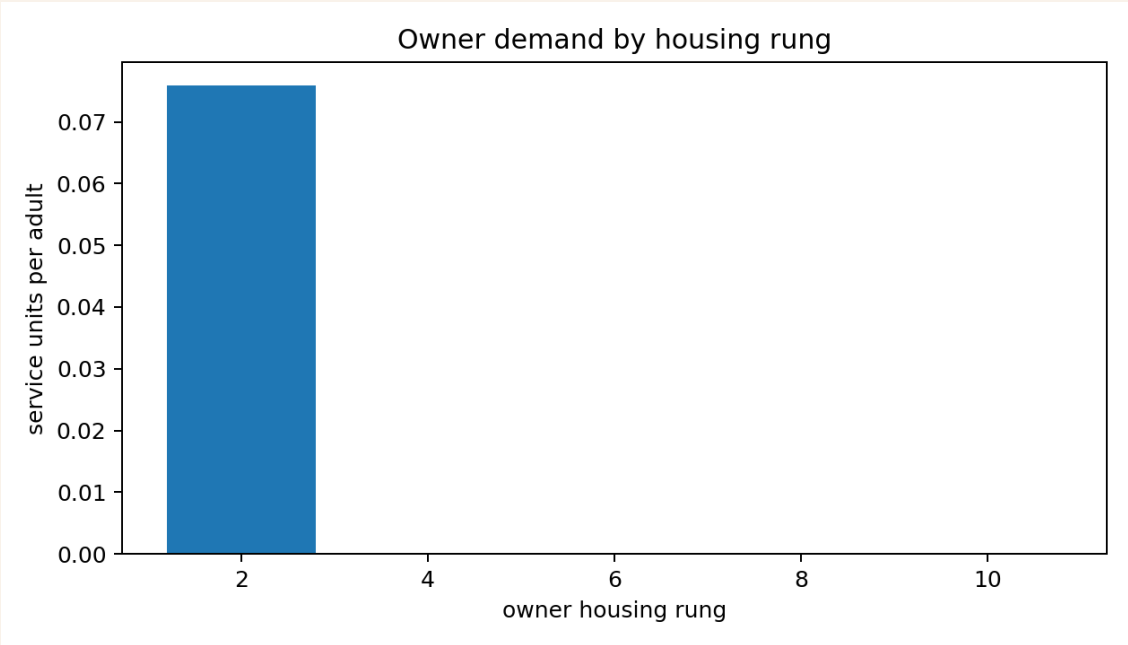
Current productivity / 100% financed



Lowest productivity / 80% financed



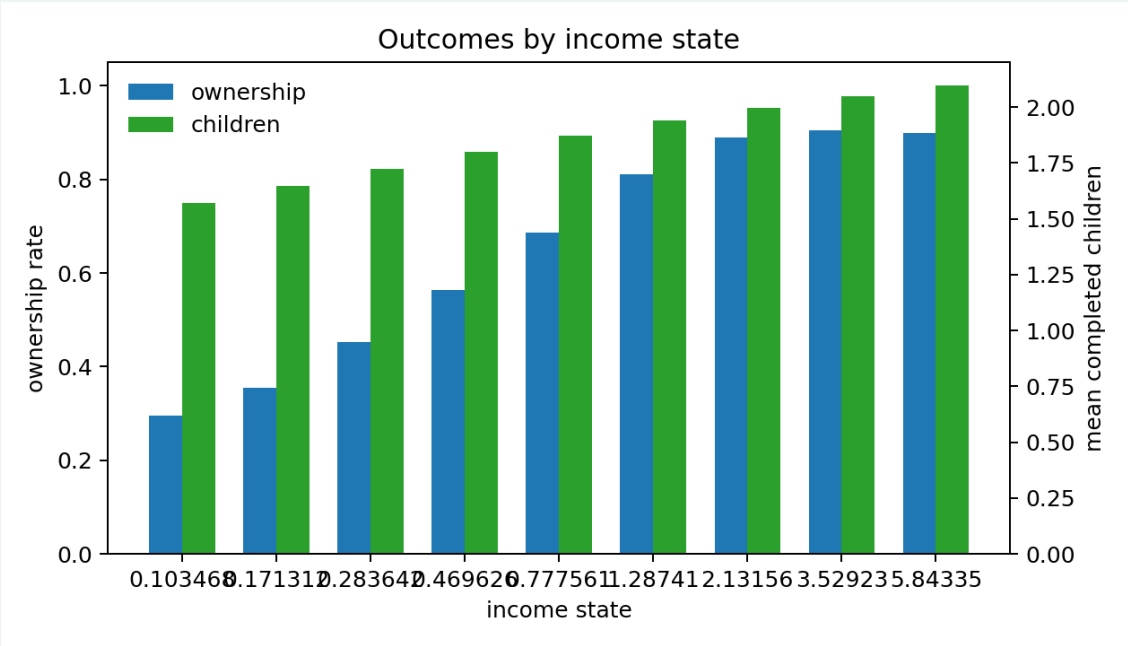
Lowest productivity / 100% financed



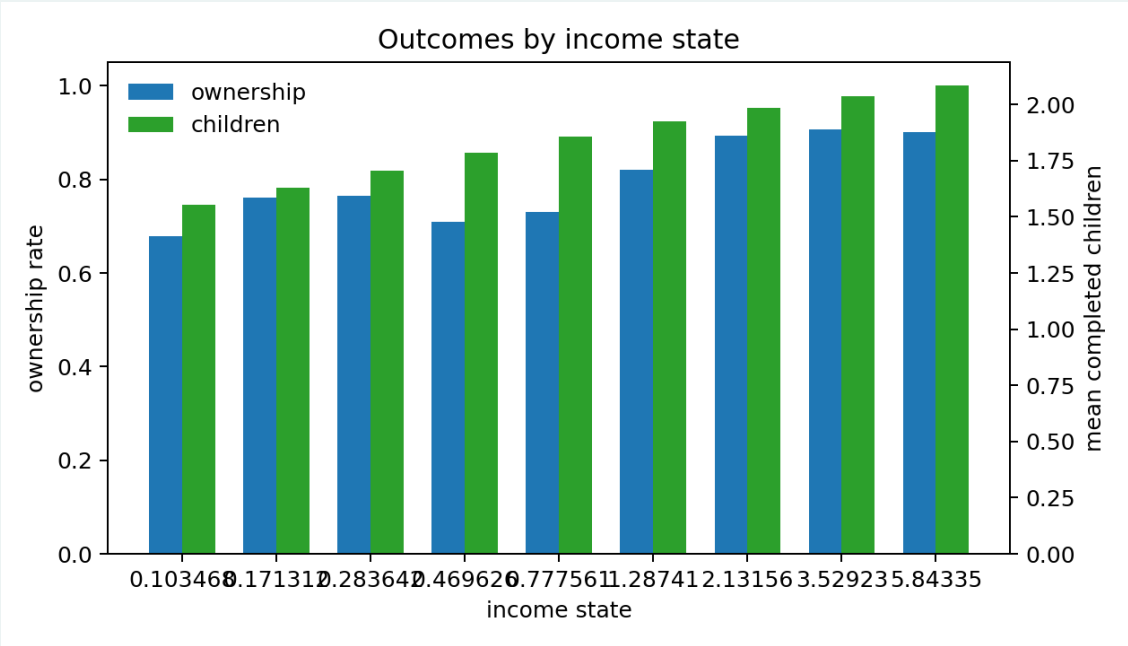
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Outcomes by income state

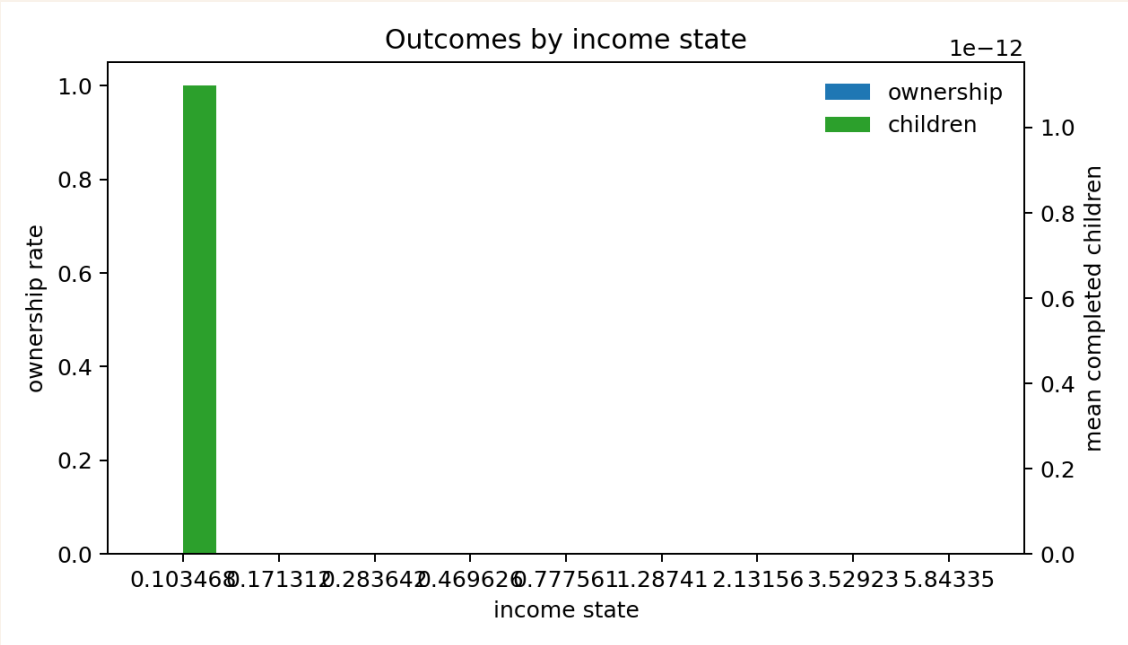
Current productivity / 80% financed



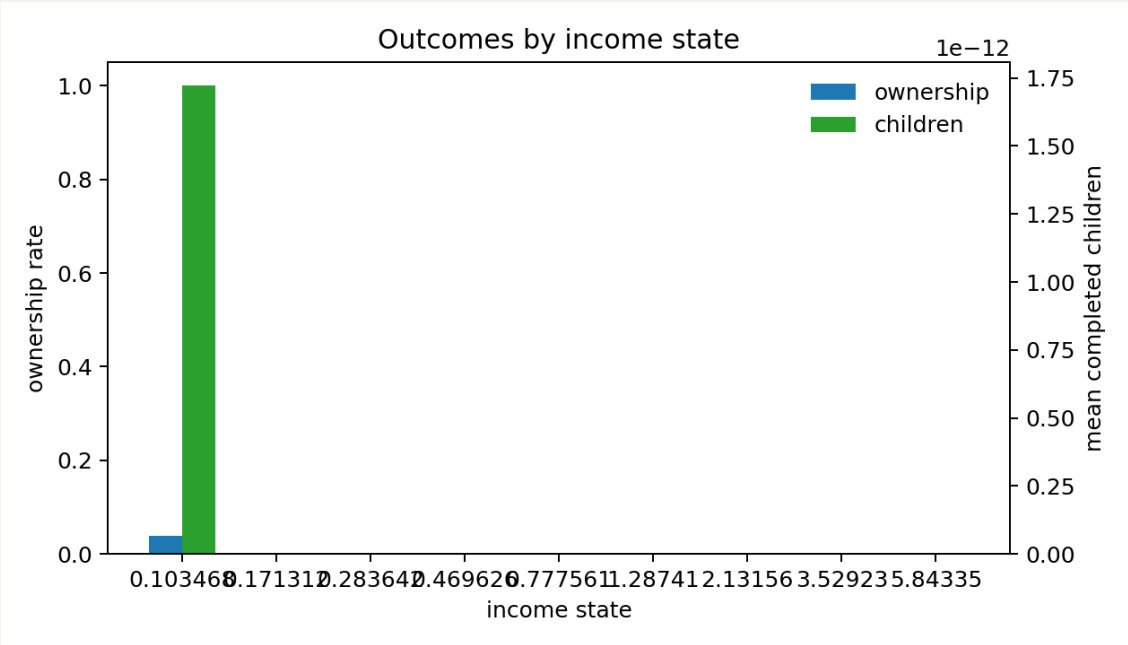
Current productivity / 100% financed



Lowest productivity / 80% financed



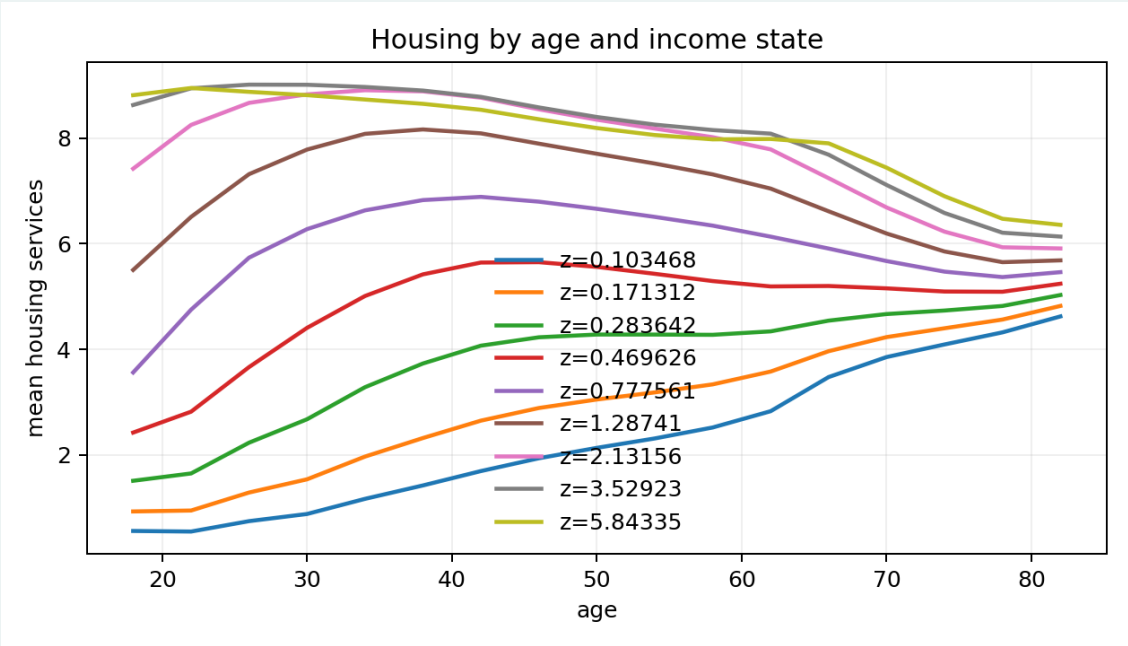
Lowest productivity / 100% financed



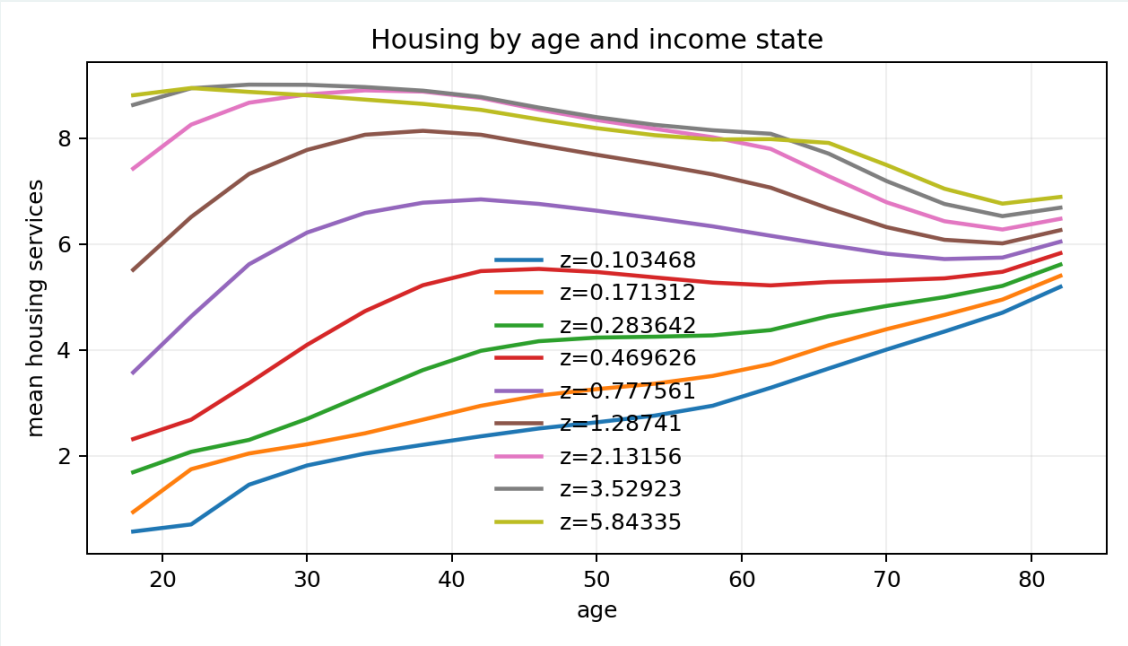
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Raw policies may show unoccupied numerical states. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Housing by age and income state

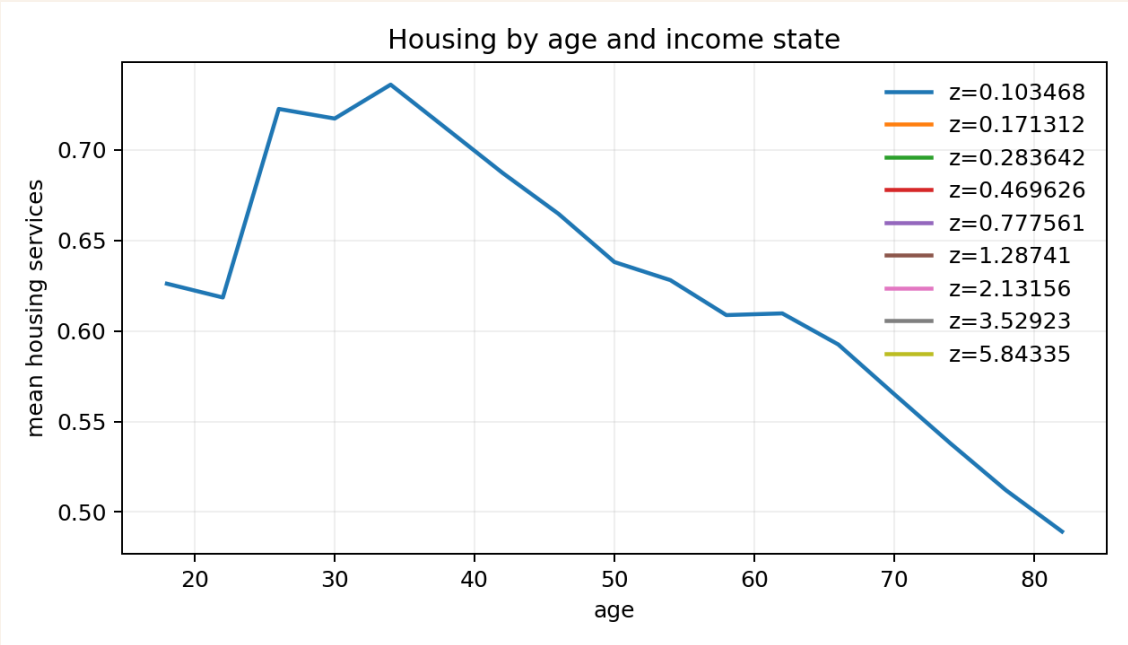
Current productivity / 80% financed



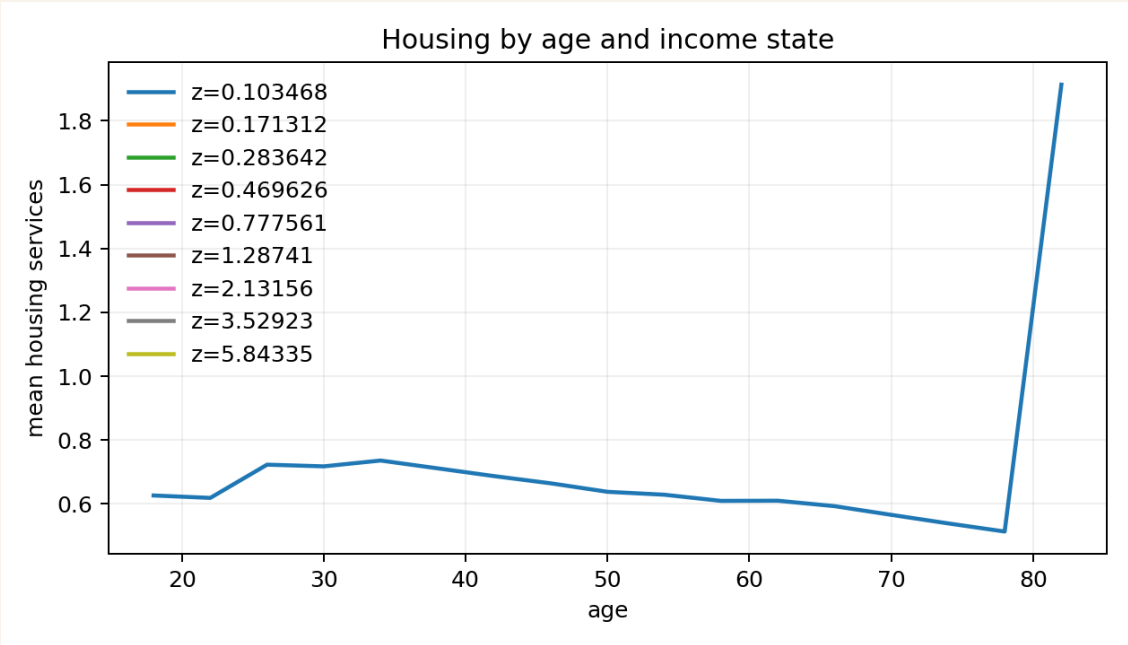
Current productivity / 100% financed



Lowest productivity / 80% financed



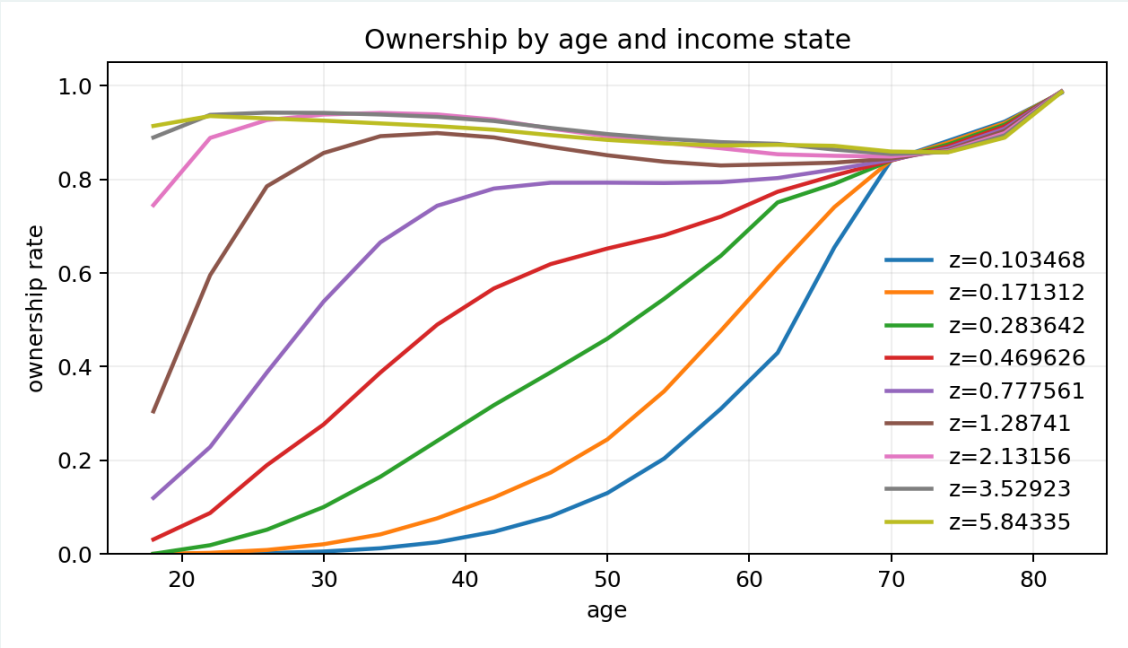
Lowest productivity / 100% financed



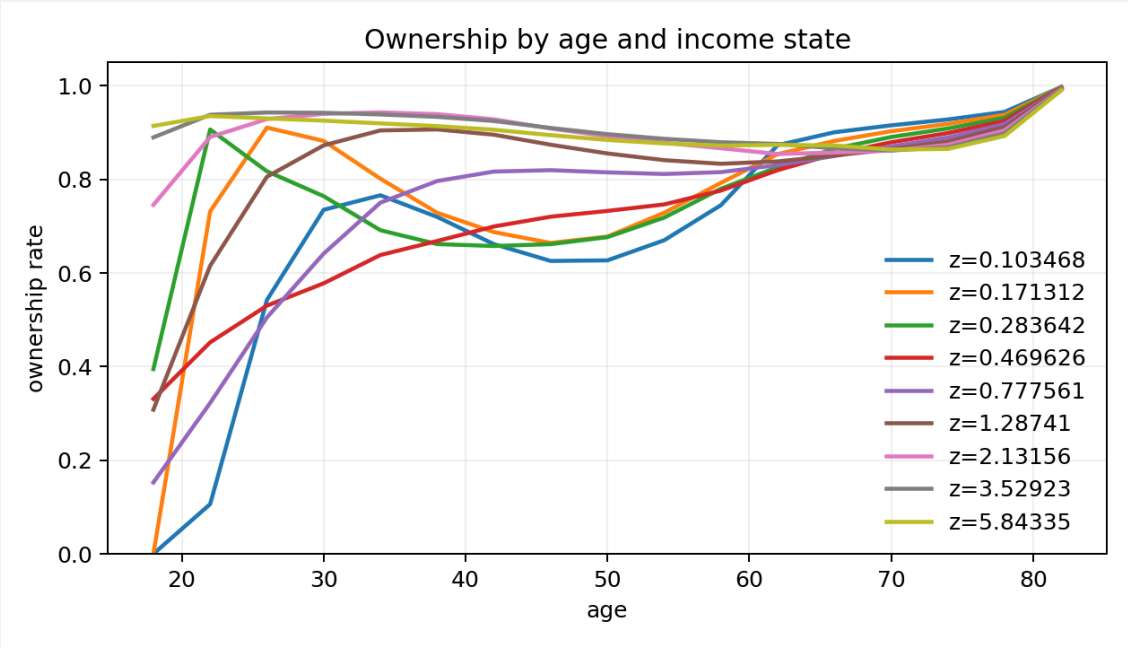
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Ownership by age and income state

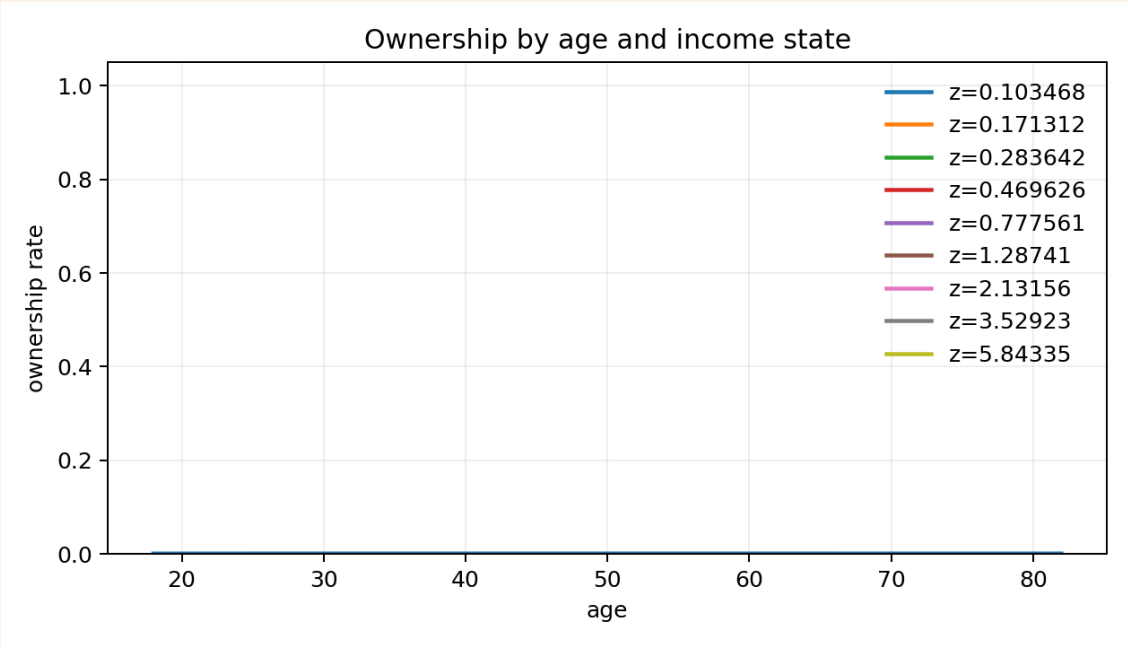
Current productivity / 80% financed



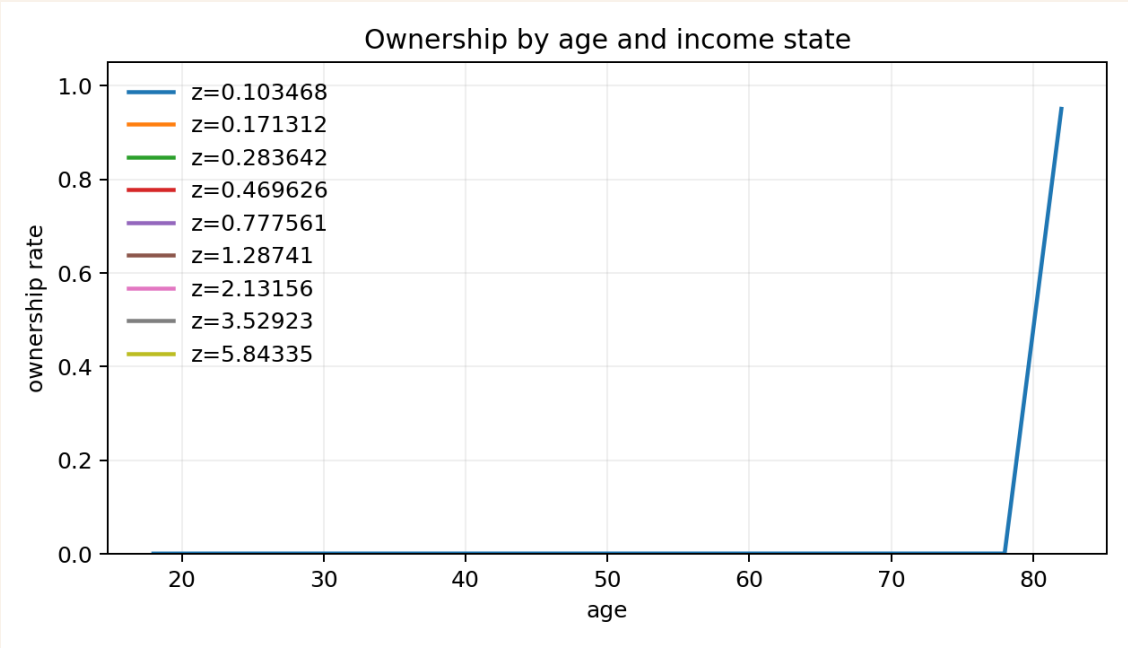
Current productivity / 100% financed



Lowest productivity / 80% financed



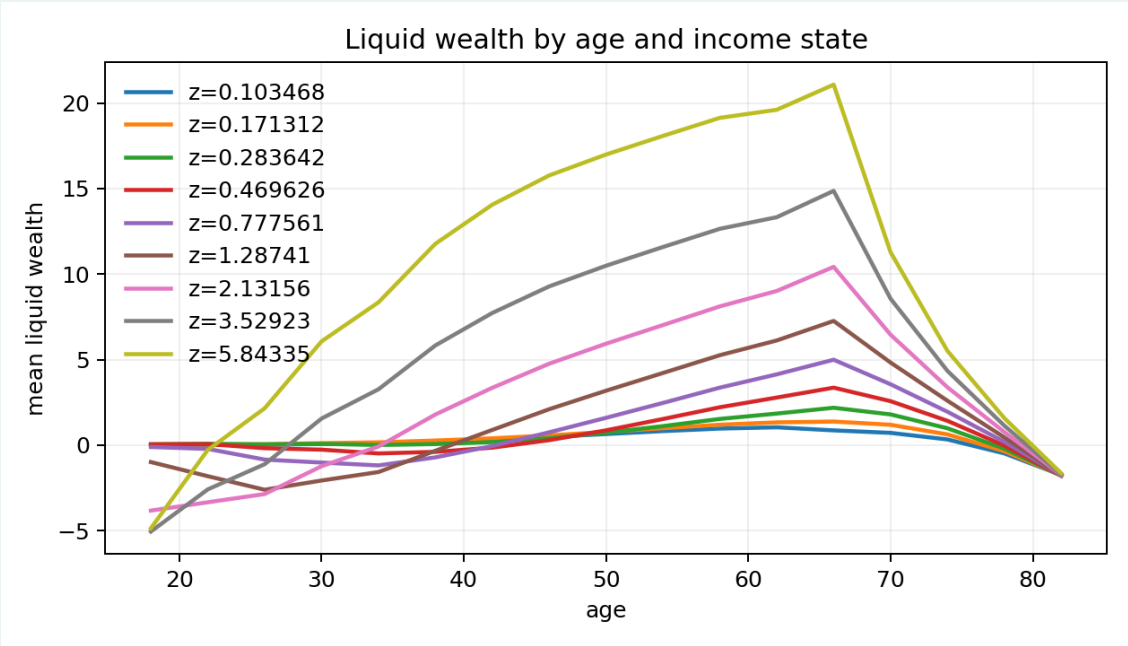
Lowest productivity / 100% financed



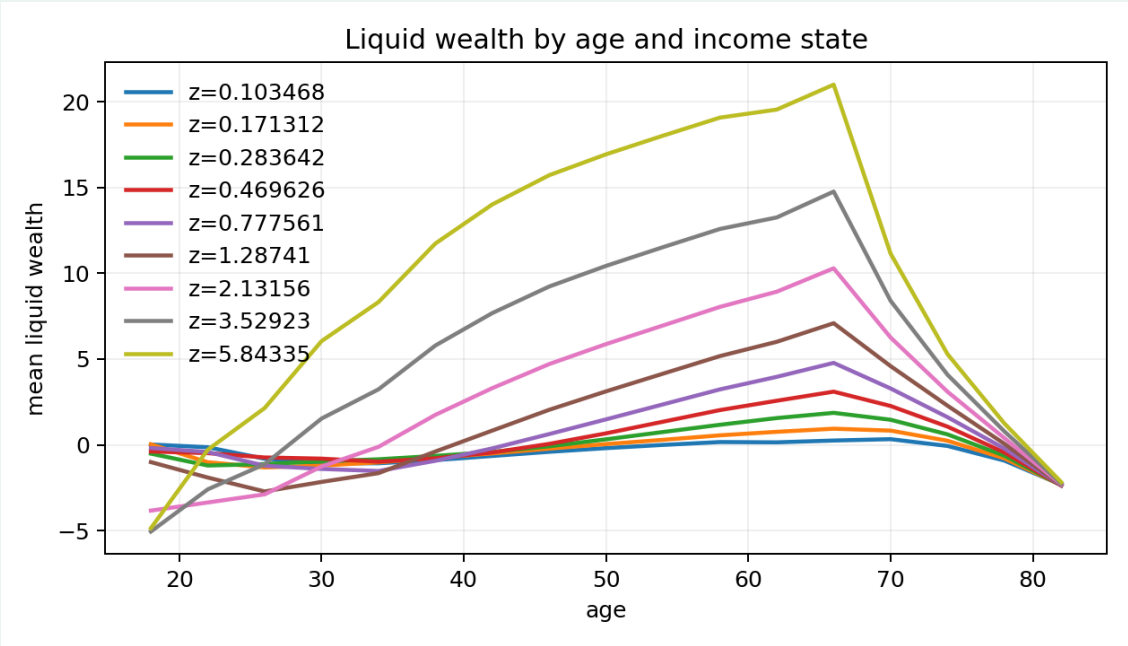
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Raw policies may show unoccupied numerical states. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Financial wealth by age and income state

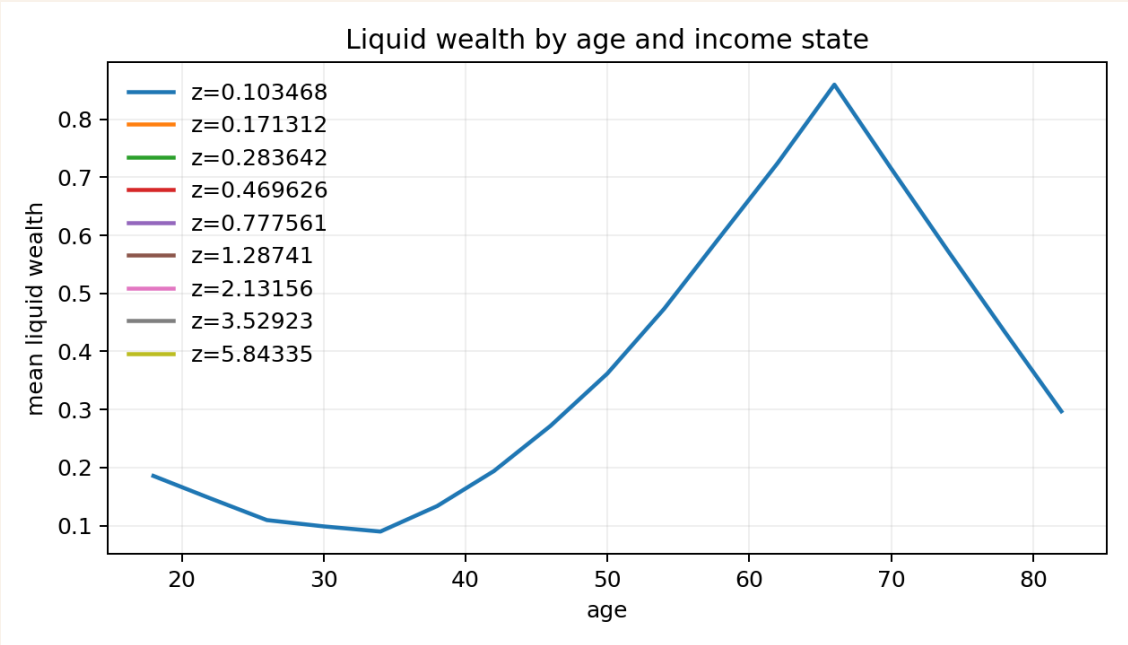
Current productivity / 80% financed



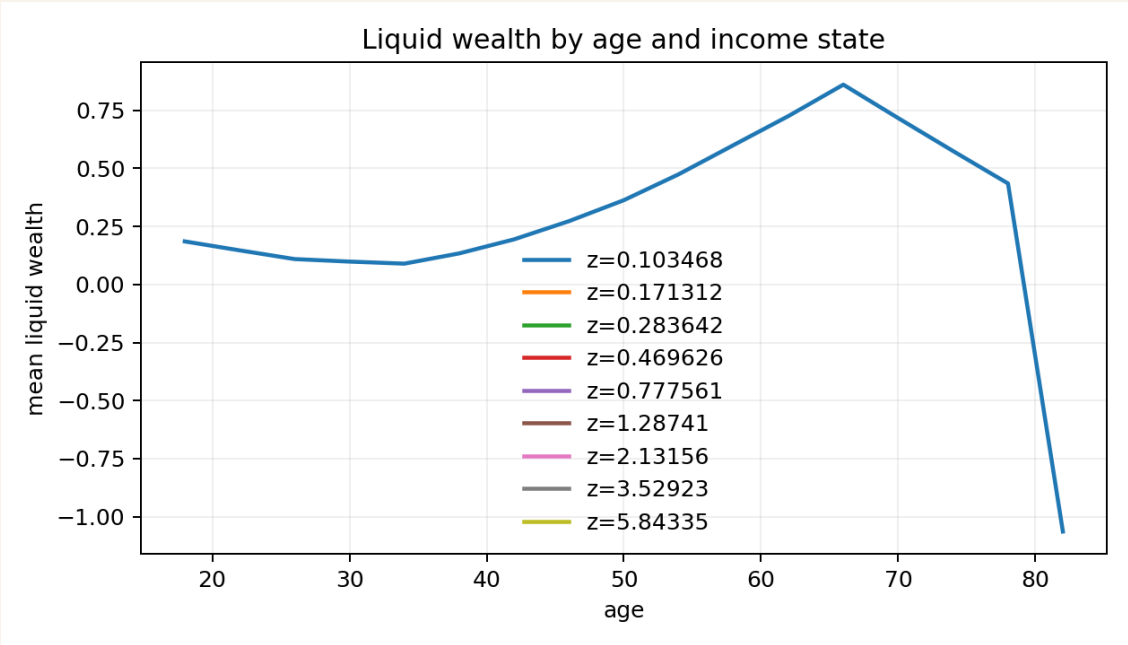
Current productivity / 100% financed



Lowest productivity / 80% financed



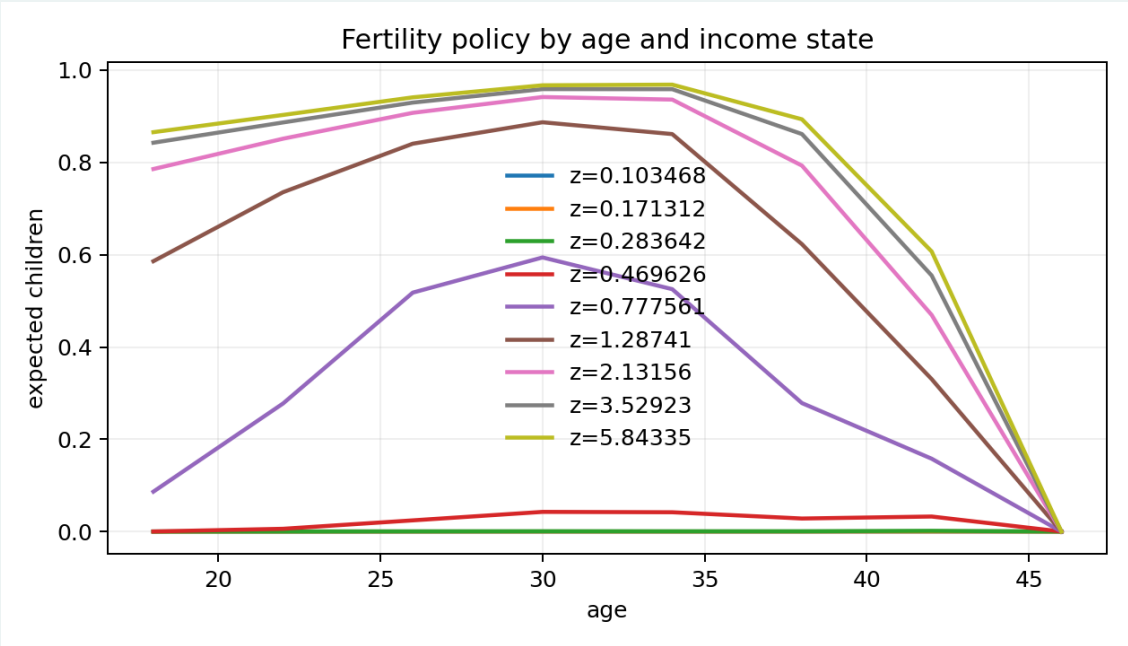
Lowest productivity / 100% financed



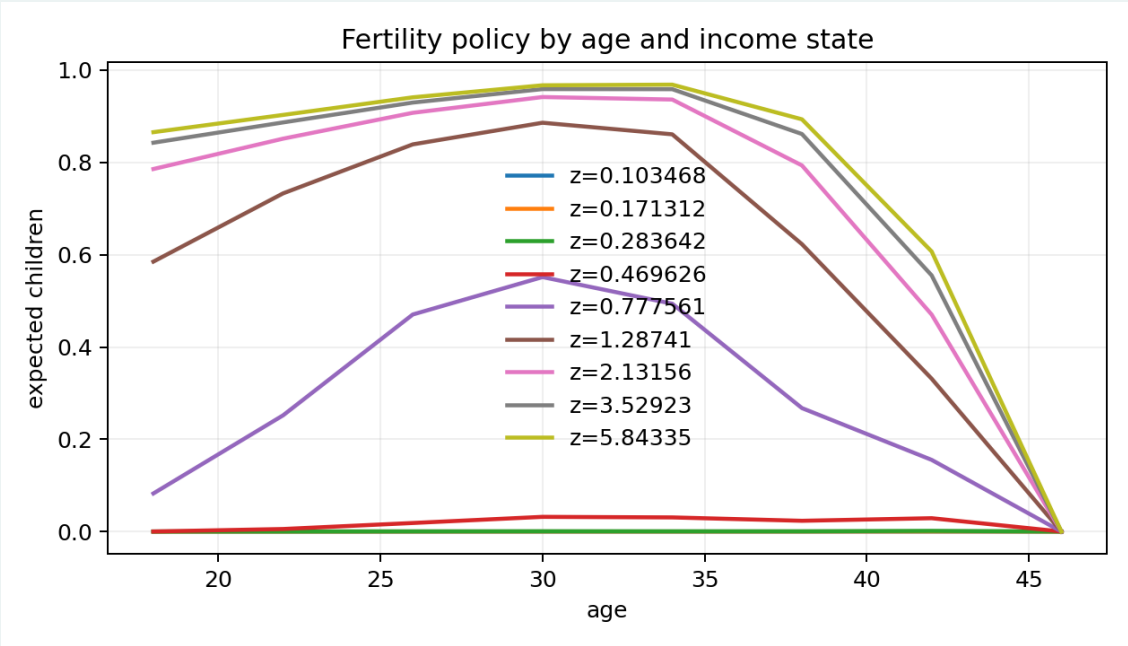
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Raw policies may show unoccupied numerical states. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Fertility choices by age and income state

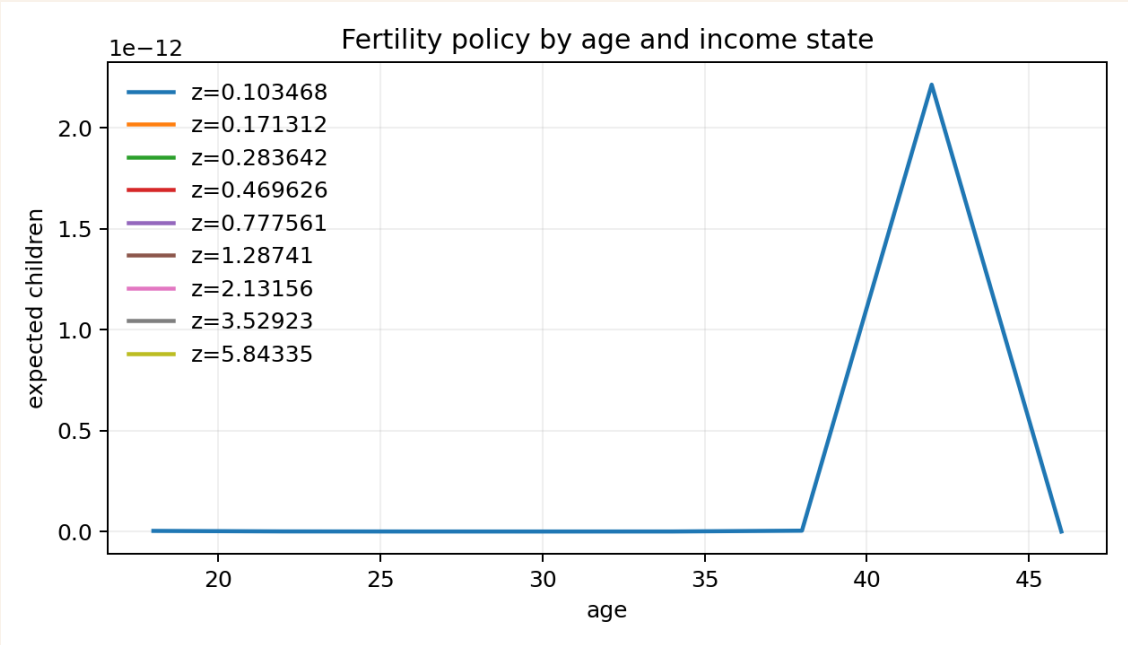
Current productivity / 80% financed



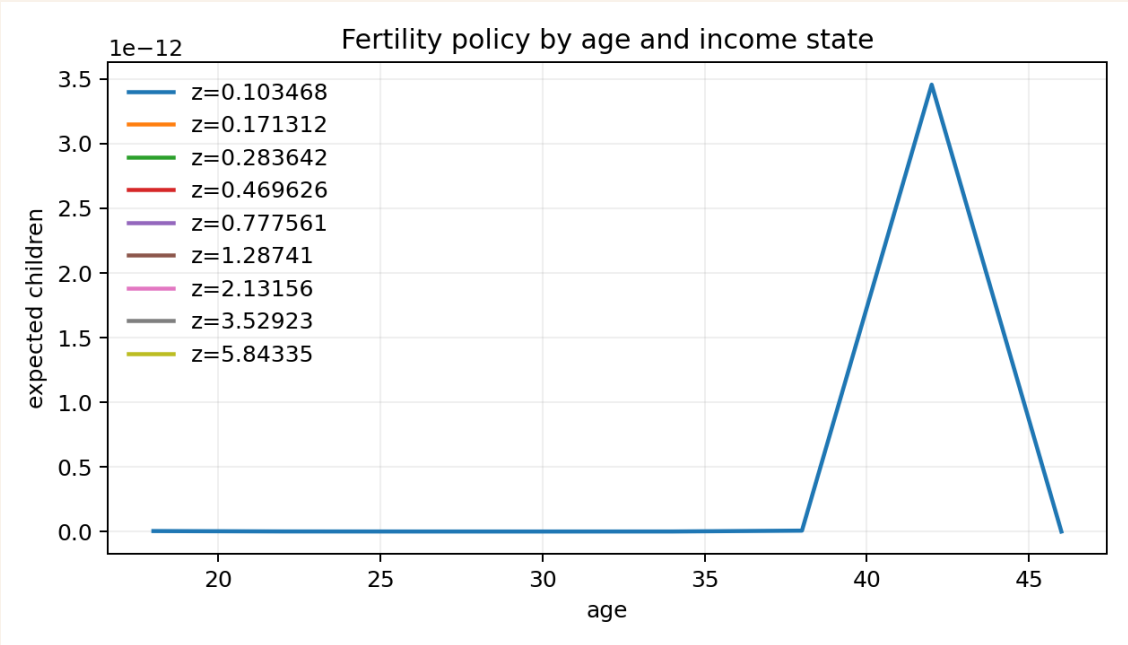
Current productivity / 100% financed



Lowest productivity / 80% financed



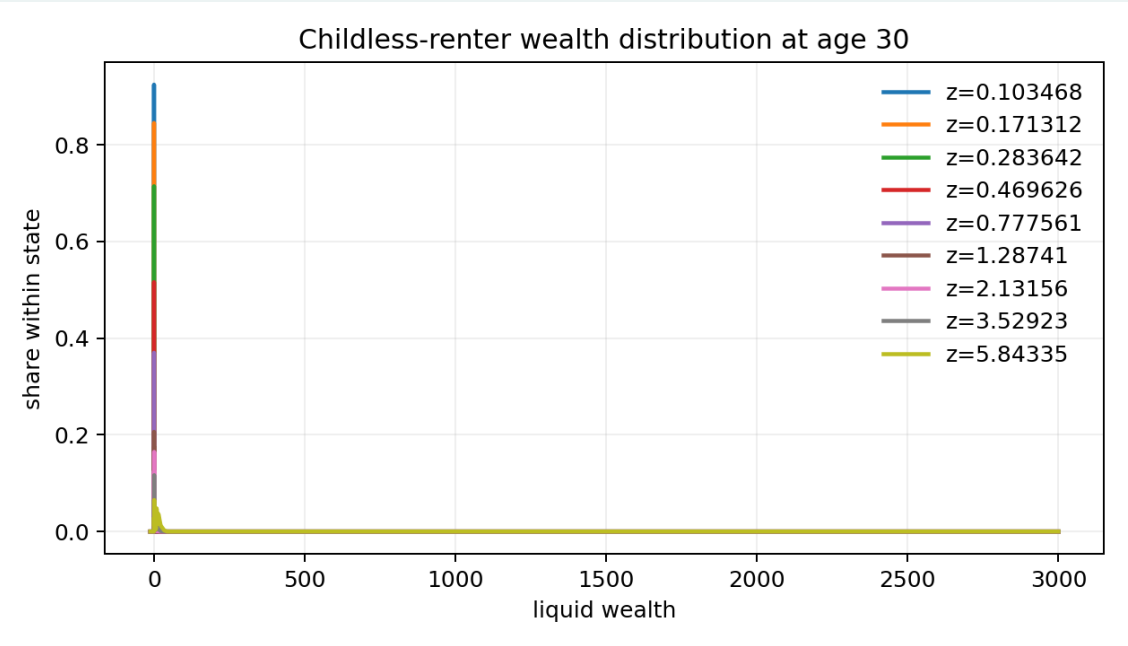
Lowest productivity / 100% financed



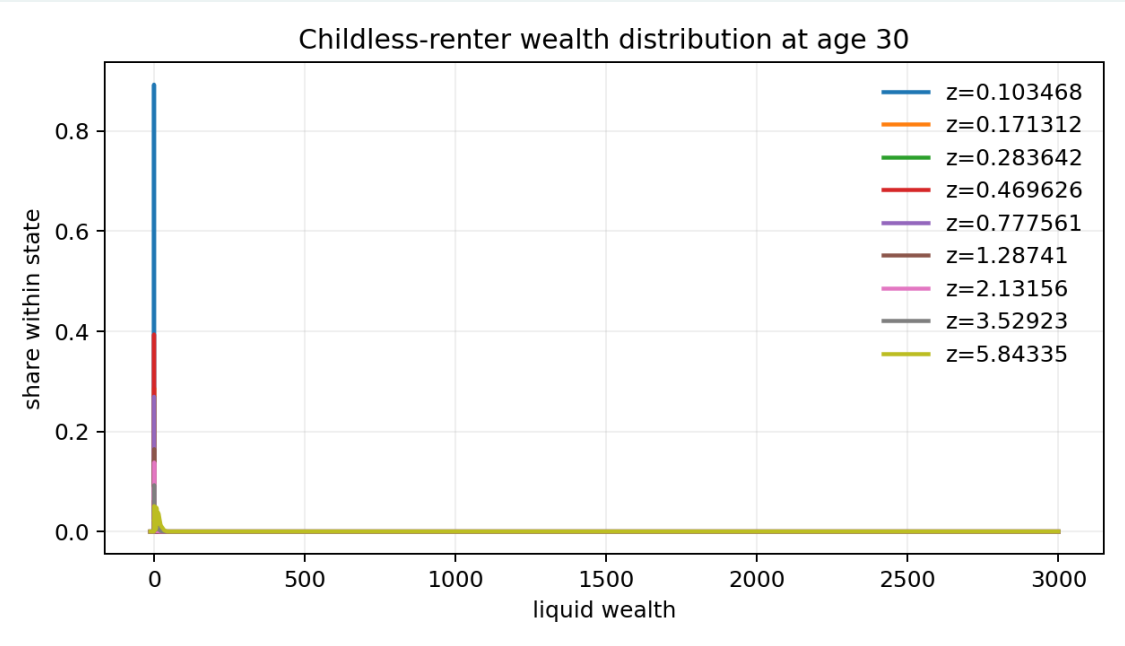
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Raw policies may show unoccupied numerical states. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Wealth: childless renters at age 30

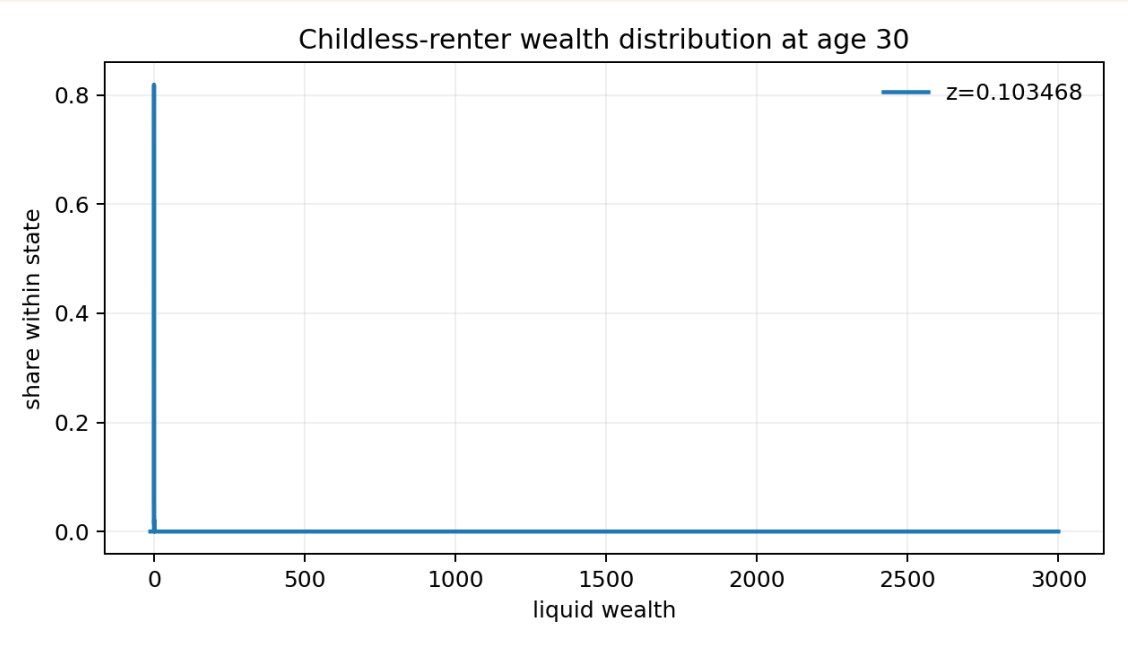
Current productivity / 80% financed



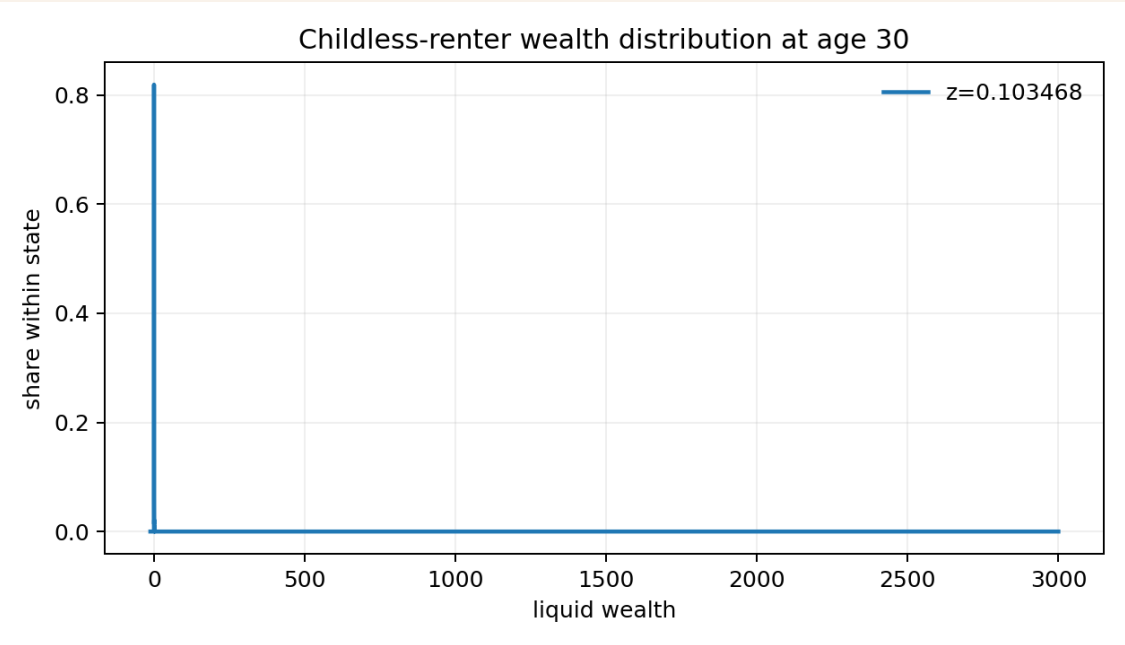
Current productivity / 100% financed



Lowest productivity / 80% financed



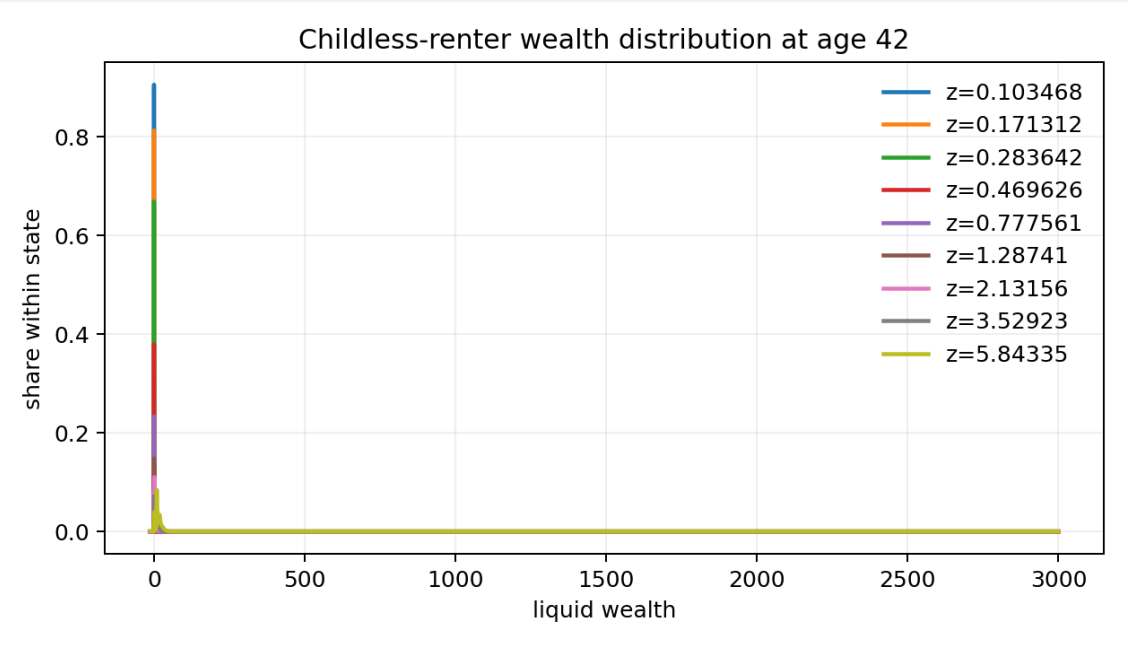
Lowest productivity / 100% financed



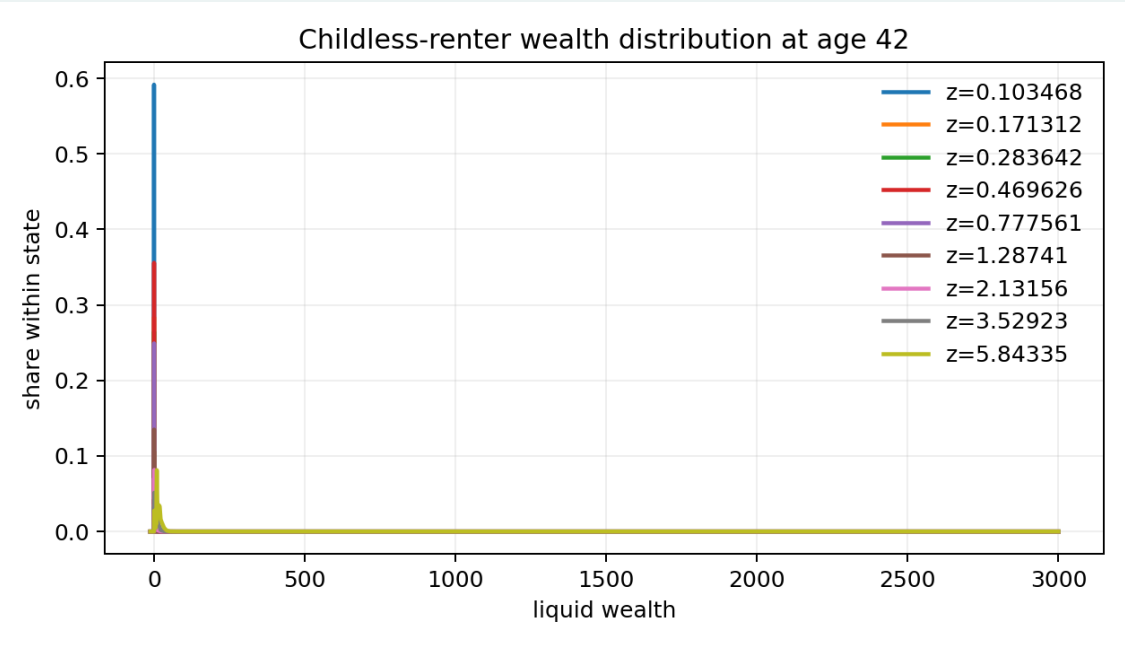
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Wealth: childless renters at age 42

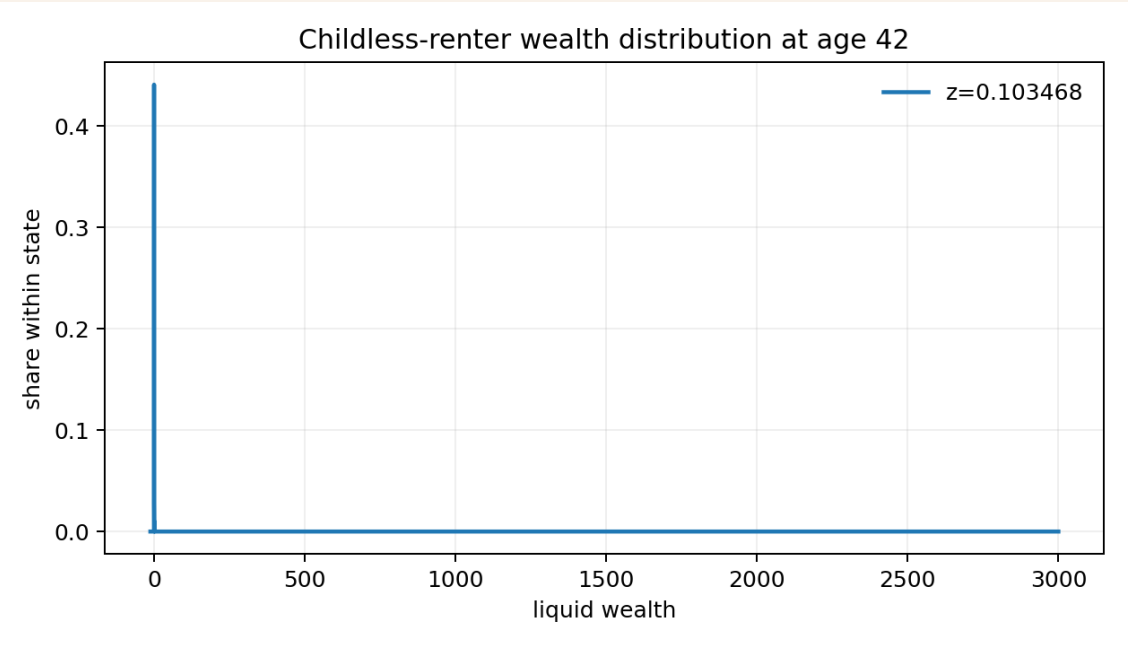
Current productivity / 80% financed



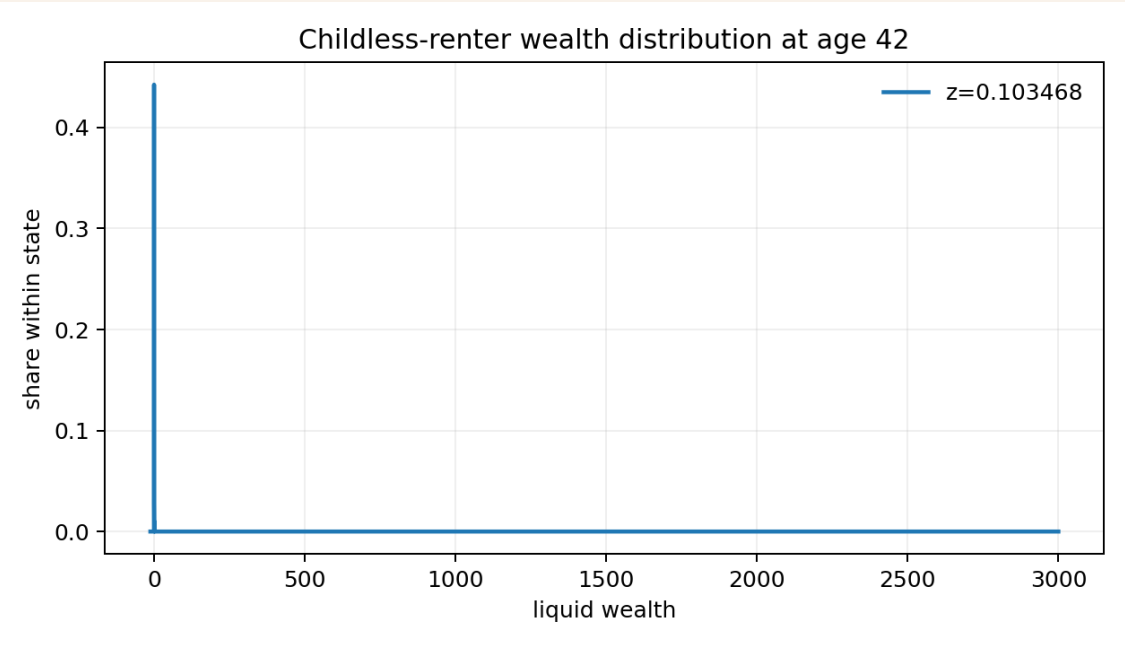
Current productivity / 100% financed



Lowest productivity / 80% financed



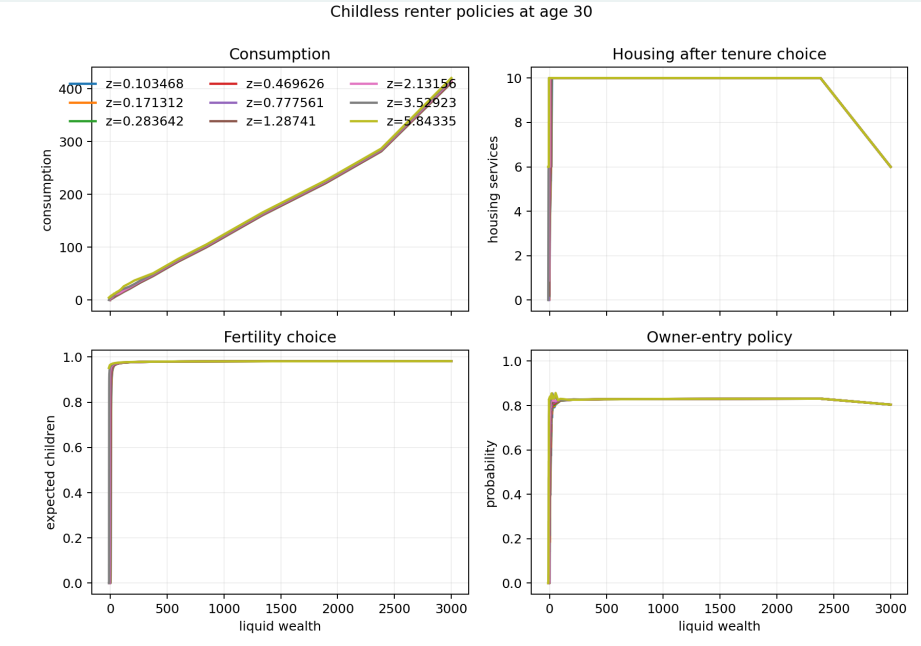
Lowest productivity / 100% financed



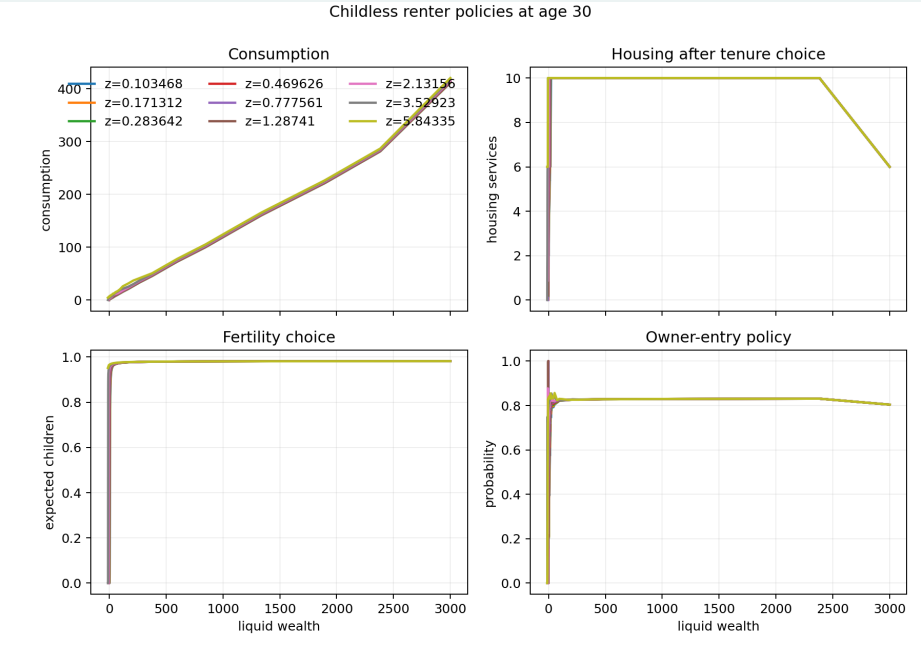
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Policies: childless renters at age 30

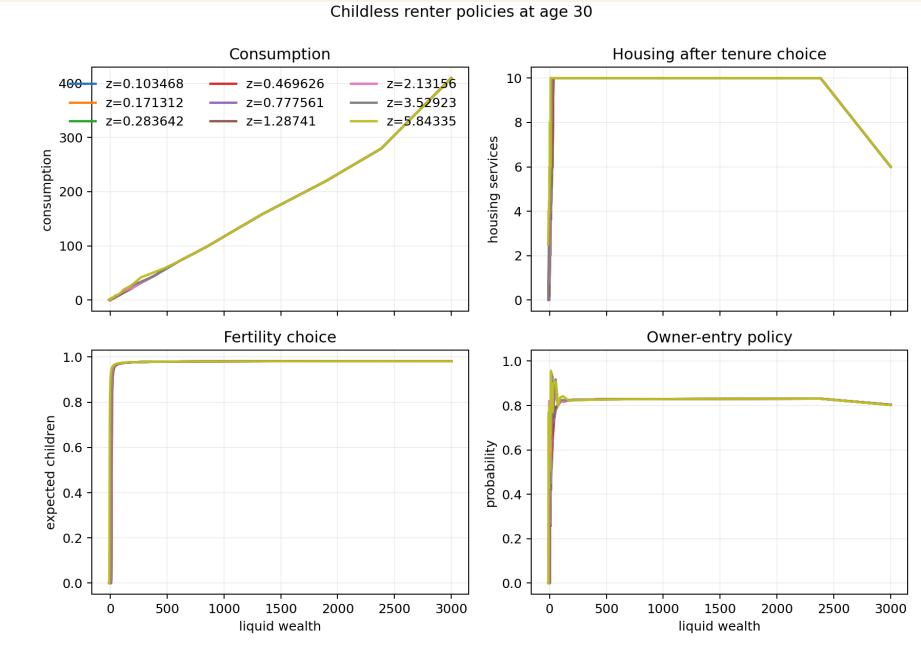
Current productivity / 80% financed



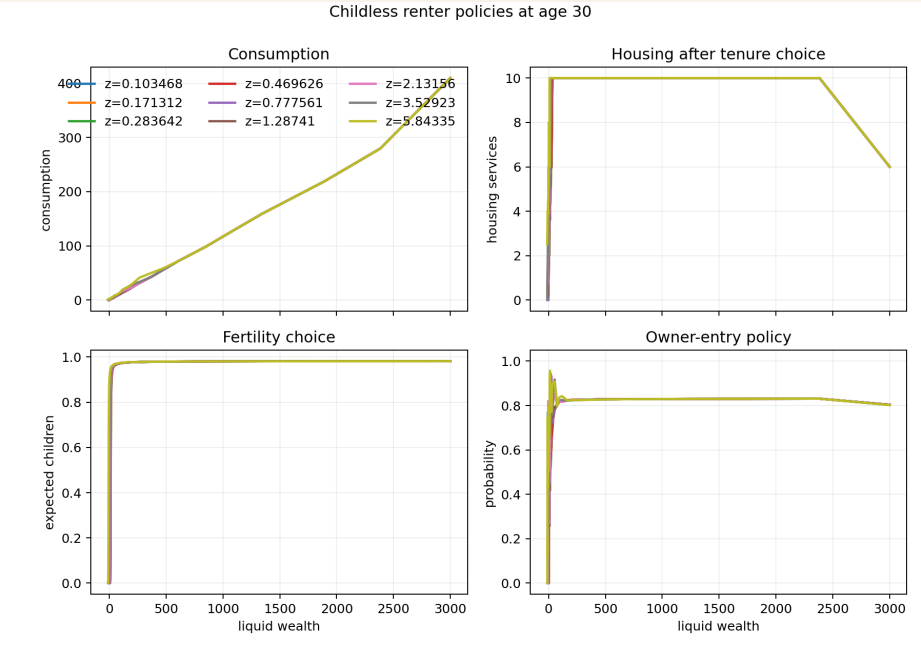
Current productivity / 100% financed



Lowest productivity / 80% financed



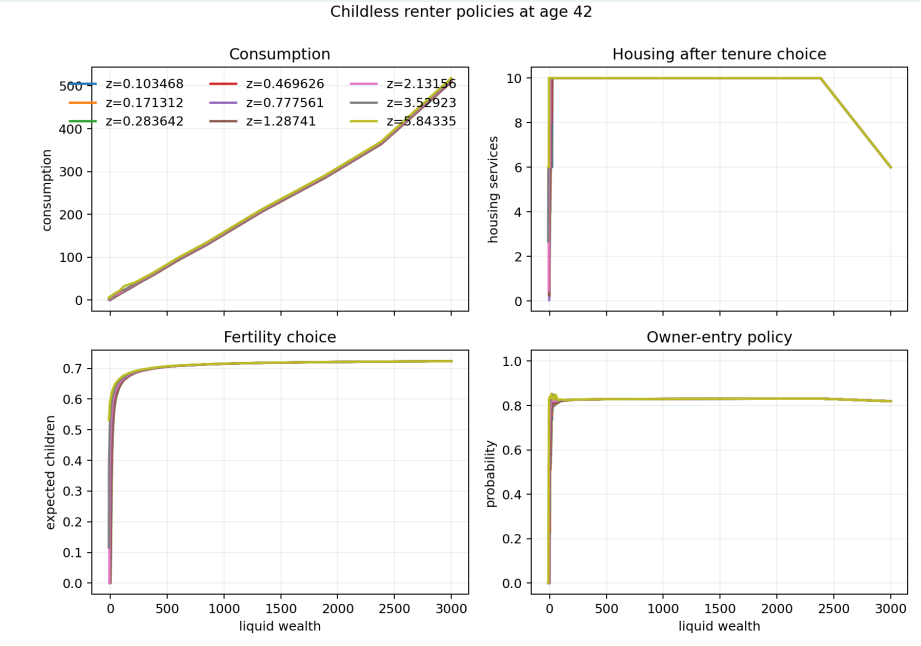
Lowest productivity / 100% financed



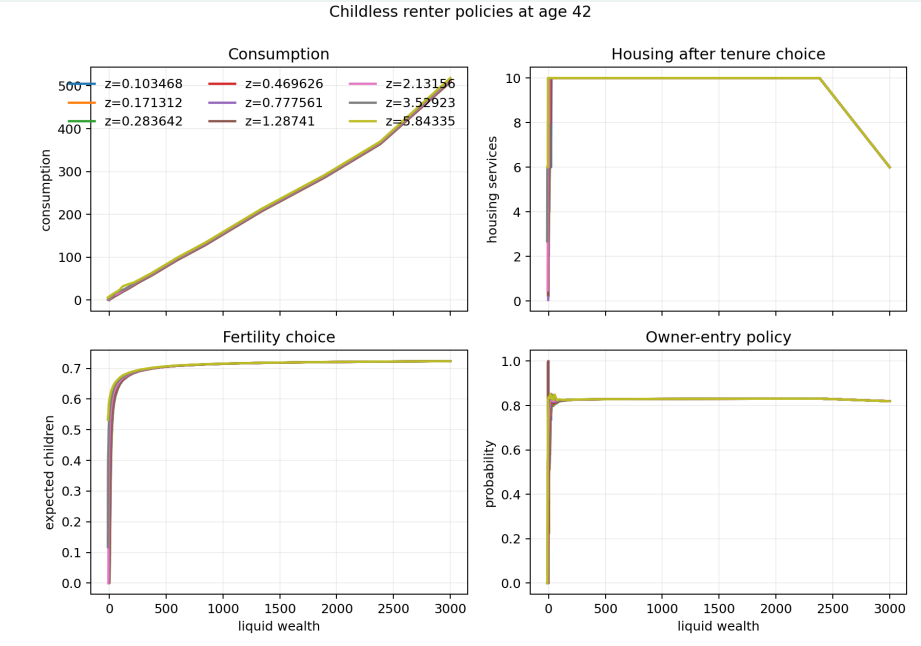
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Raw policies may show unoccupied numerical states. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Policies: childless renters at age 42

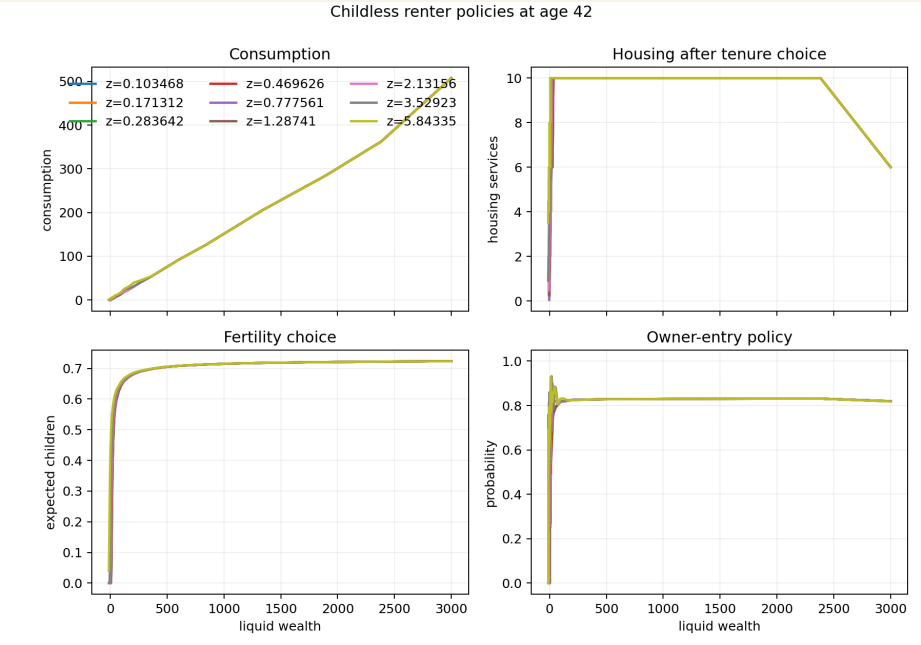
Current productivity / 80% financed



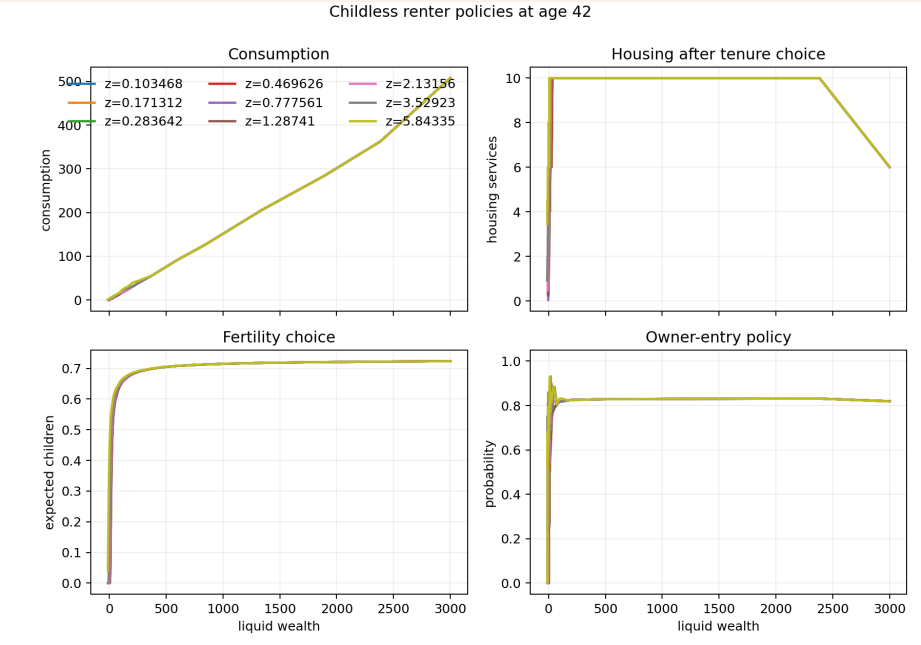
Current productivity / 100% financed



Lowest productivity / 80% financed



Lowest productivity / 100% financed



Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Raw policies may show unoccupied numerical states. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Held housing prices

Current productivity / 80% financed



Current productivity / 100% financed



Lowest productivity / 80% financed



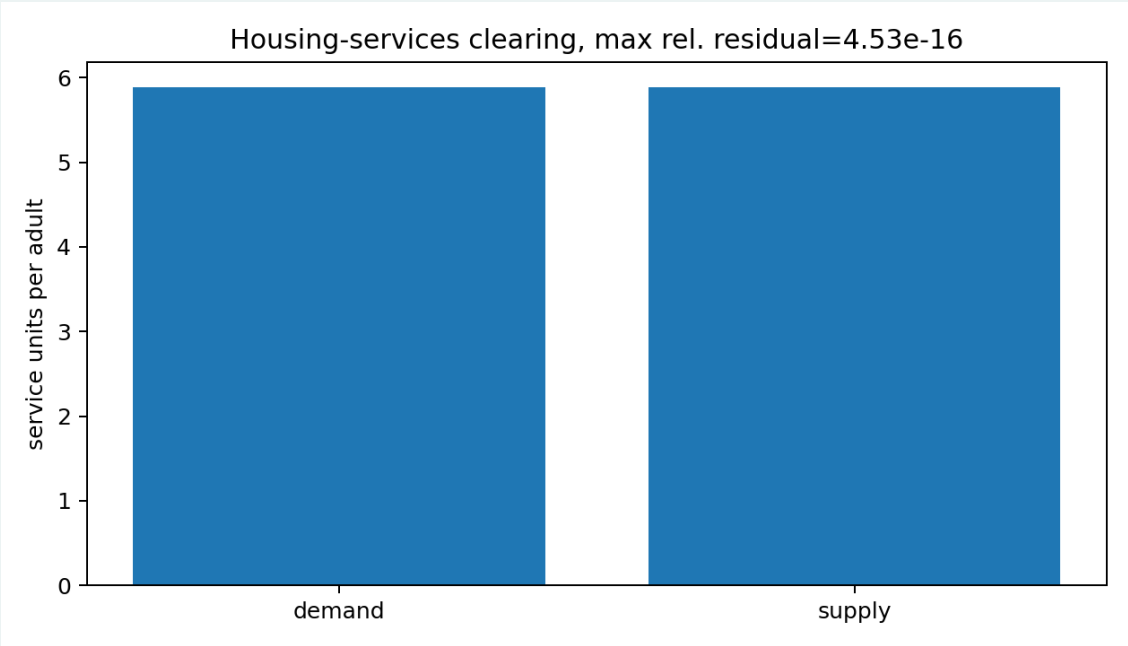
Lowest productivity / 100% financed



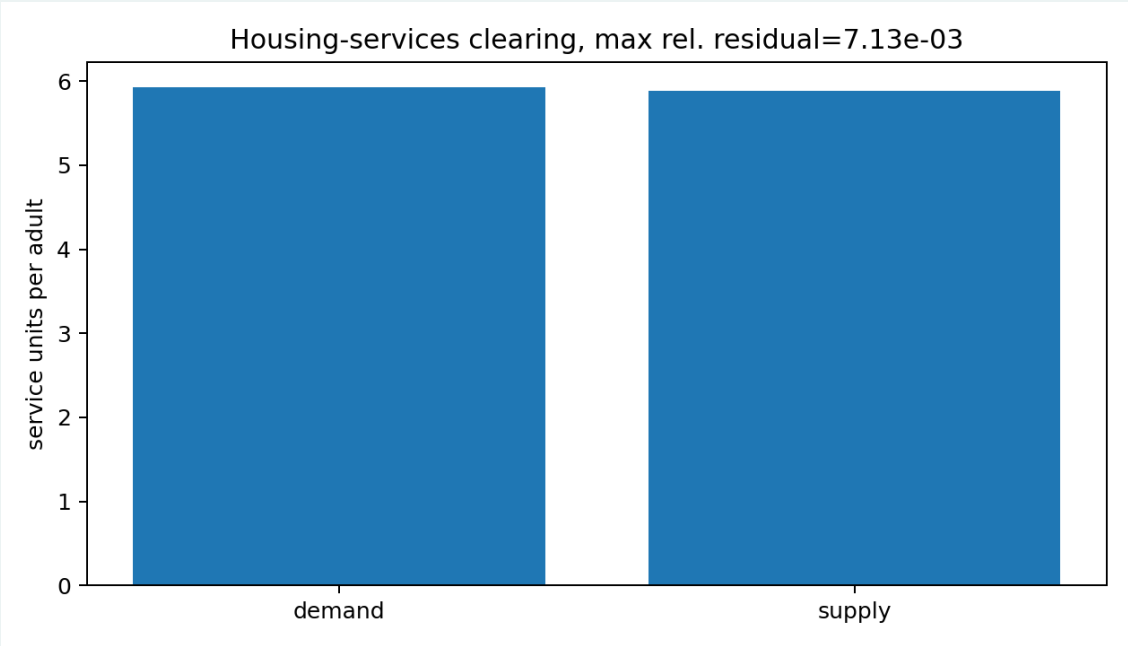
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Housing quantities at the held price

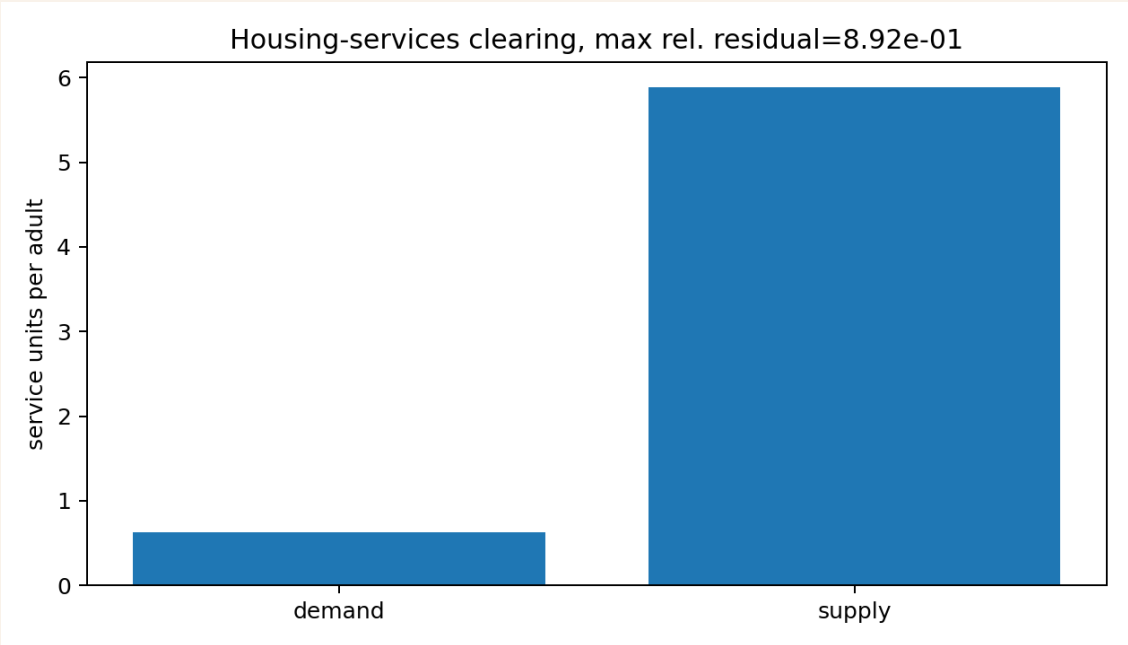
Current productivity / 80% financed



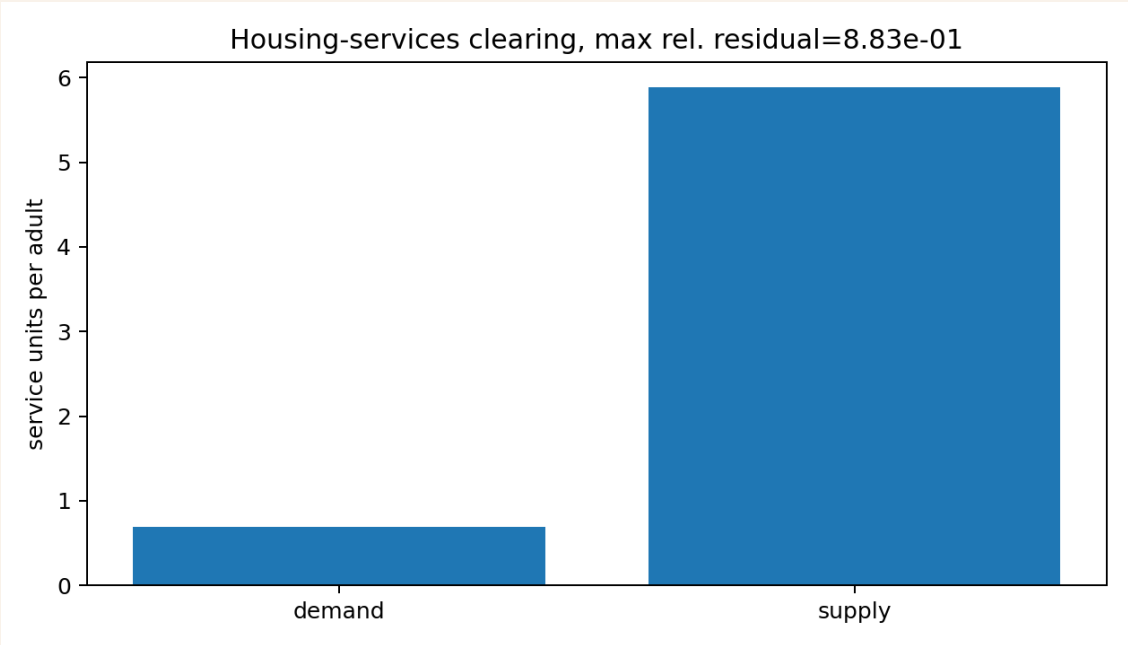
Current productivity / 100% financed



Lowest productivity / 80% financed



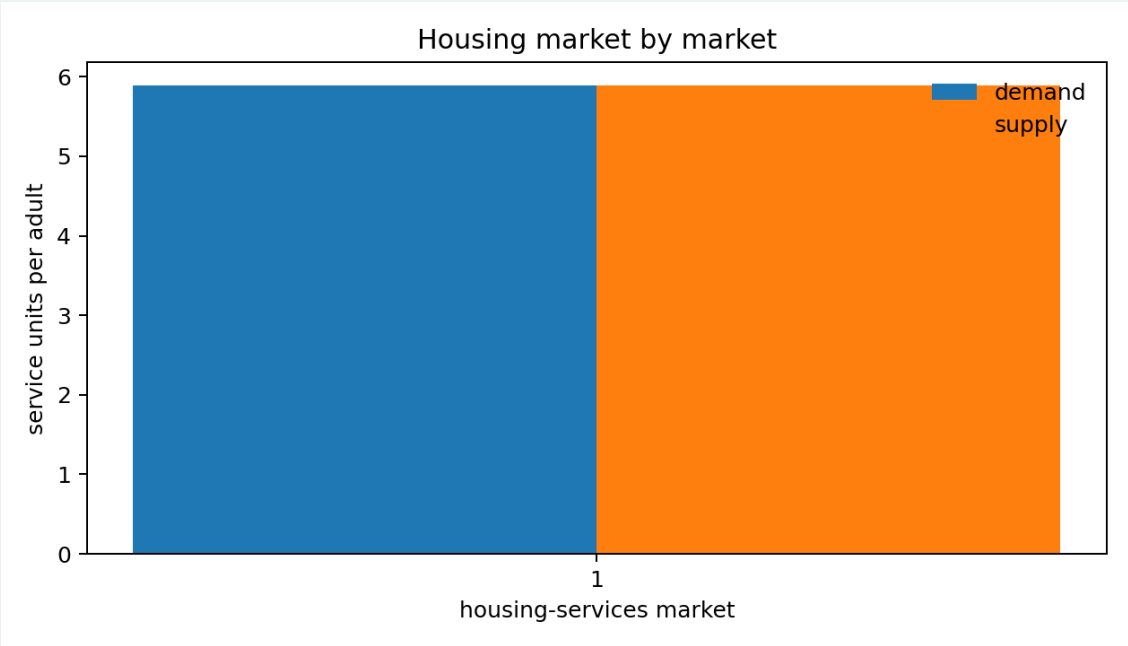
Lowest productivity / 100% financed



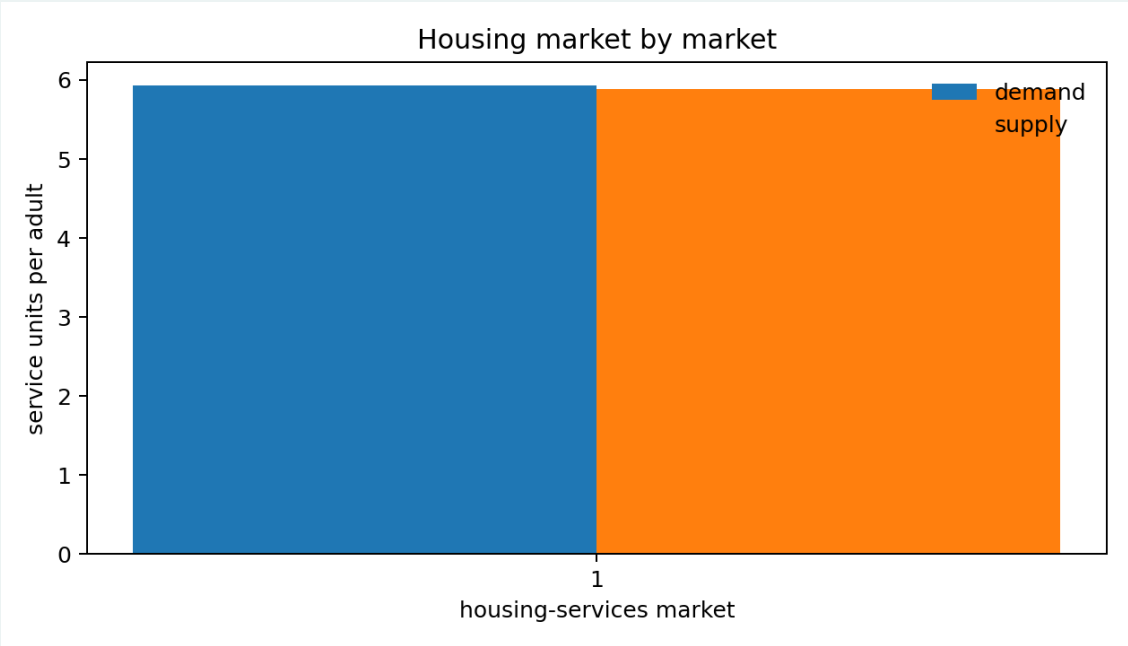
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Market residuals by market

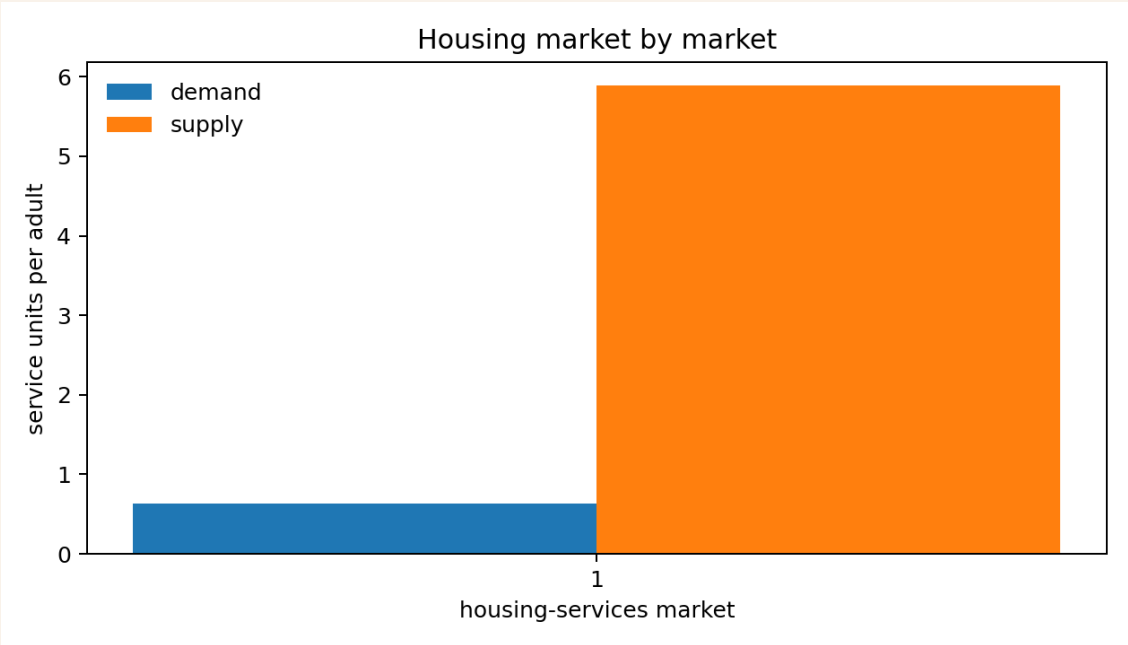
Current productivity / 80% financed



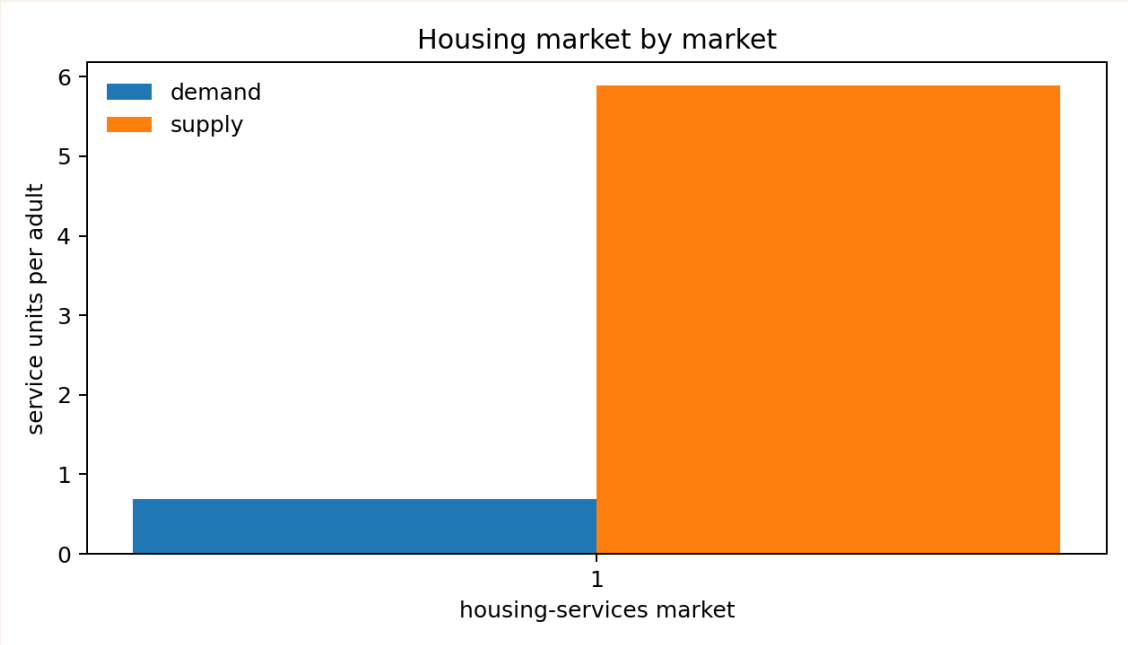
Current productivity / 100% financed



Lowest productivity / 80% financed



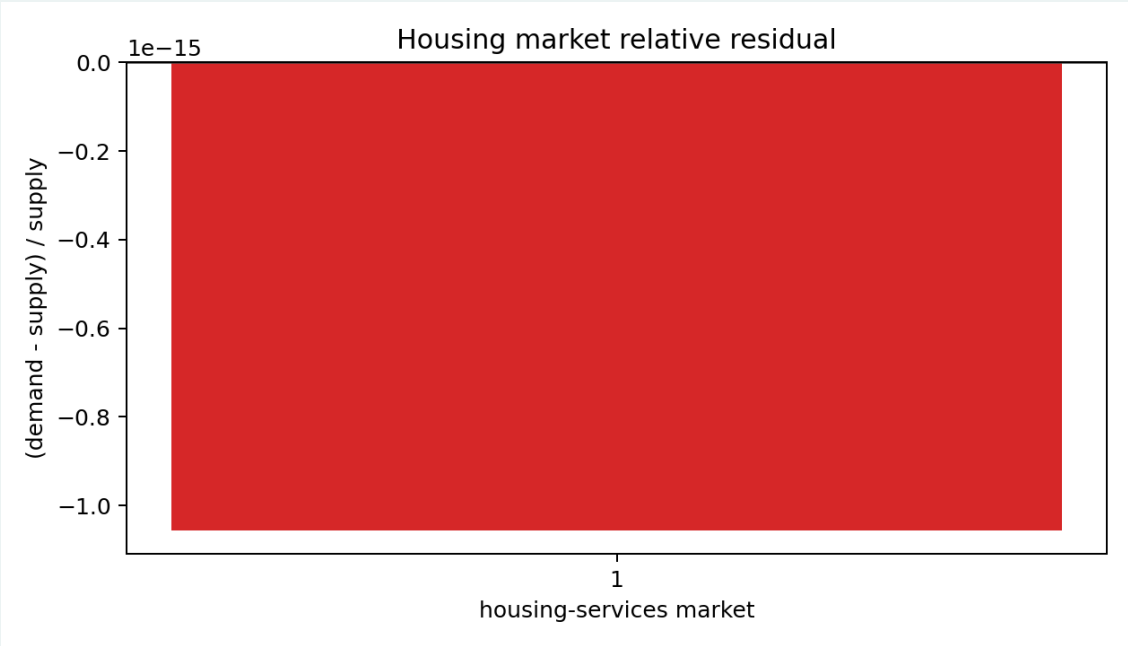
Lowest productivity / 100% financed



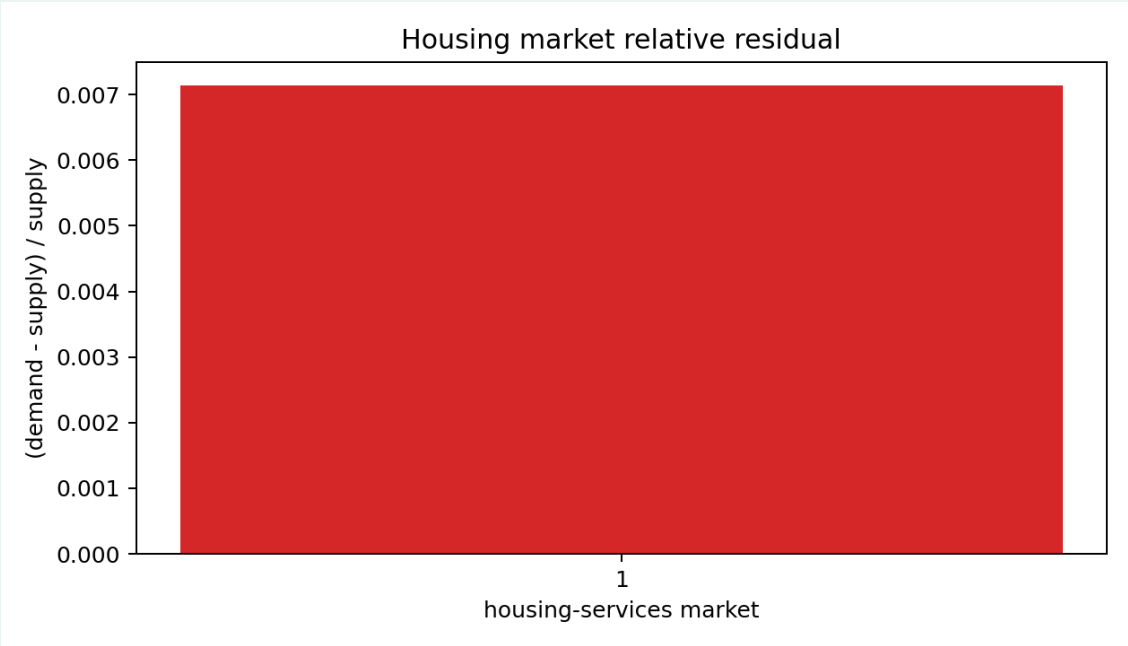
Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.

Market residuals

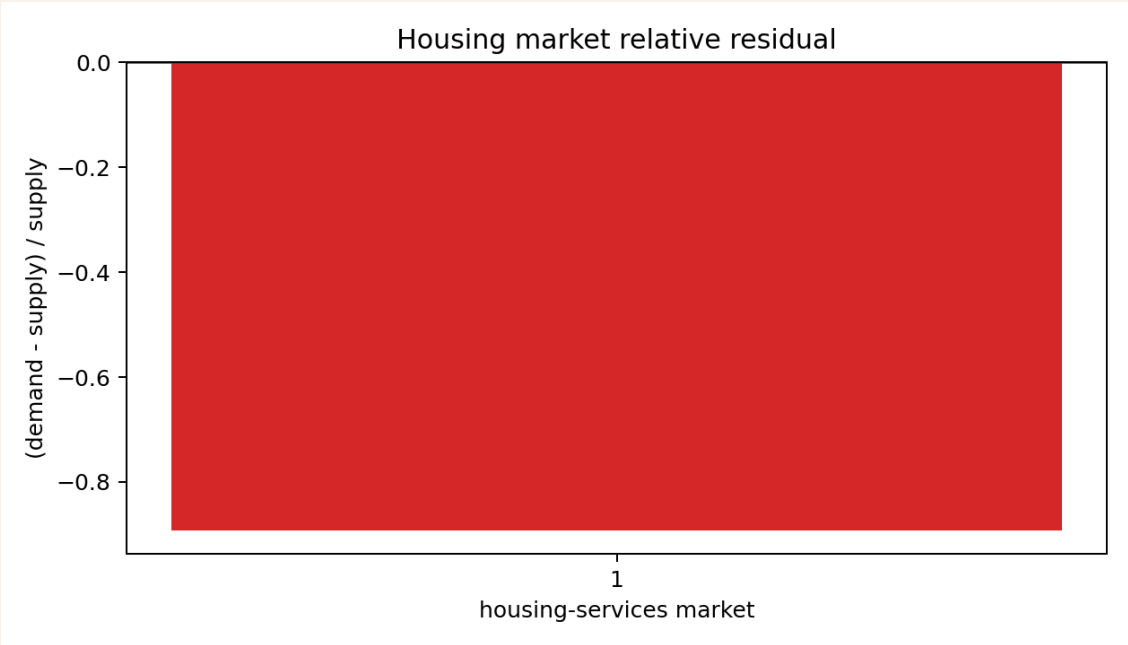
Current productivity / 80% financed



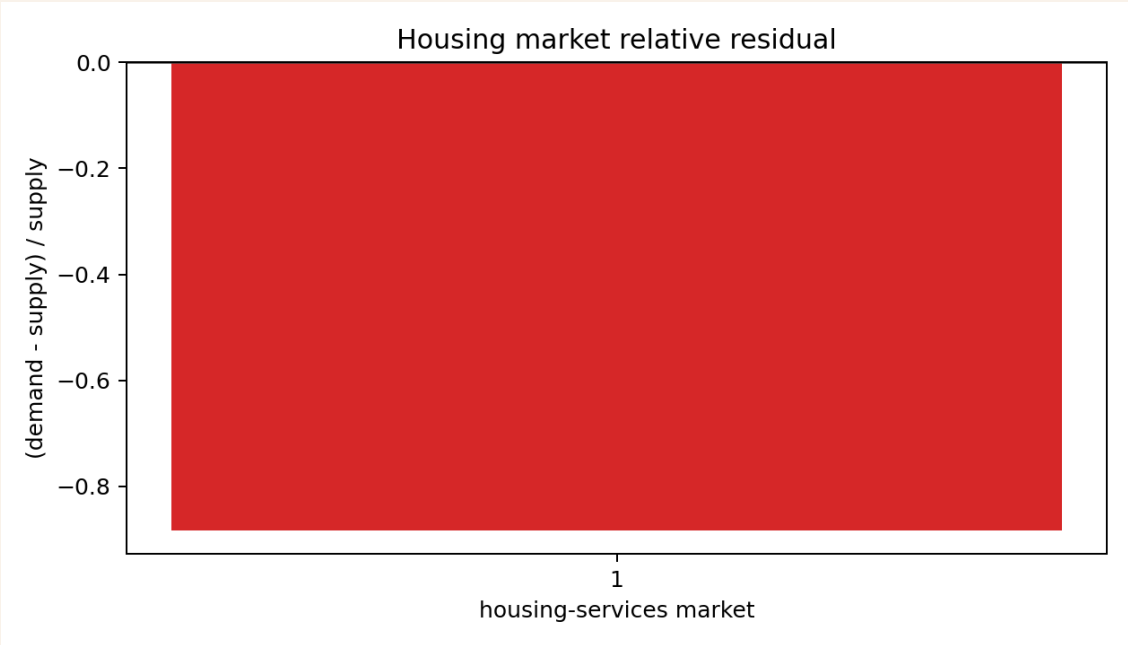
Current productivity / 100% financed



Lowest productivity / 80% financed



Lowest productivity / 100% financed



Fixed-price diagnostic; plotted market residuals are not equilibrium clearing tests. Both 100% aggregate policy reporters failed their unchanged value gate; raw diagnostics are shown, not certified aggregate consumption or saving.